Genital system

- It consists of → 1- Male
  → 2- female

Female genital system

- It consists of two ovaries, 2 uterine tubes ,uterus , vagina and 2 vulval lips.

Female genital system

1. Two Ovaries.

2. 2 Uterine tubes=fallopian tube=oviduct, Each uterine tubes:

a_ Infundibulum. b- Ampulla. c- Isthmus.

3. Uterus: it consists of the following parts:

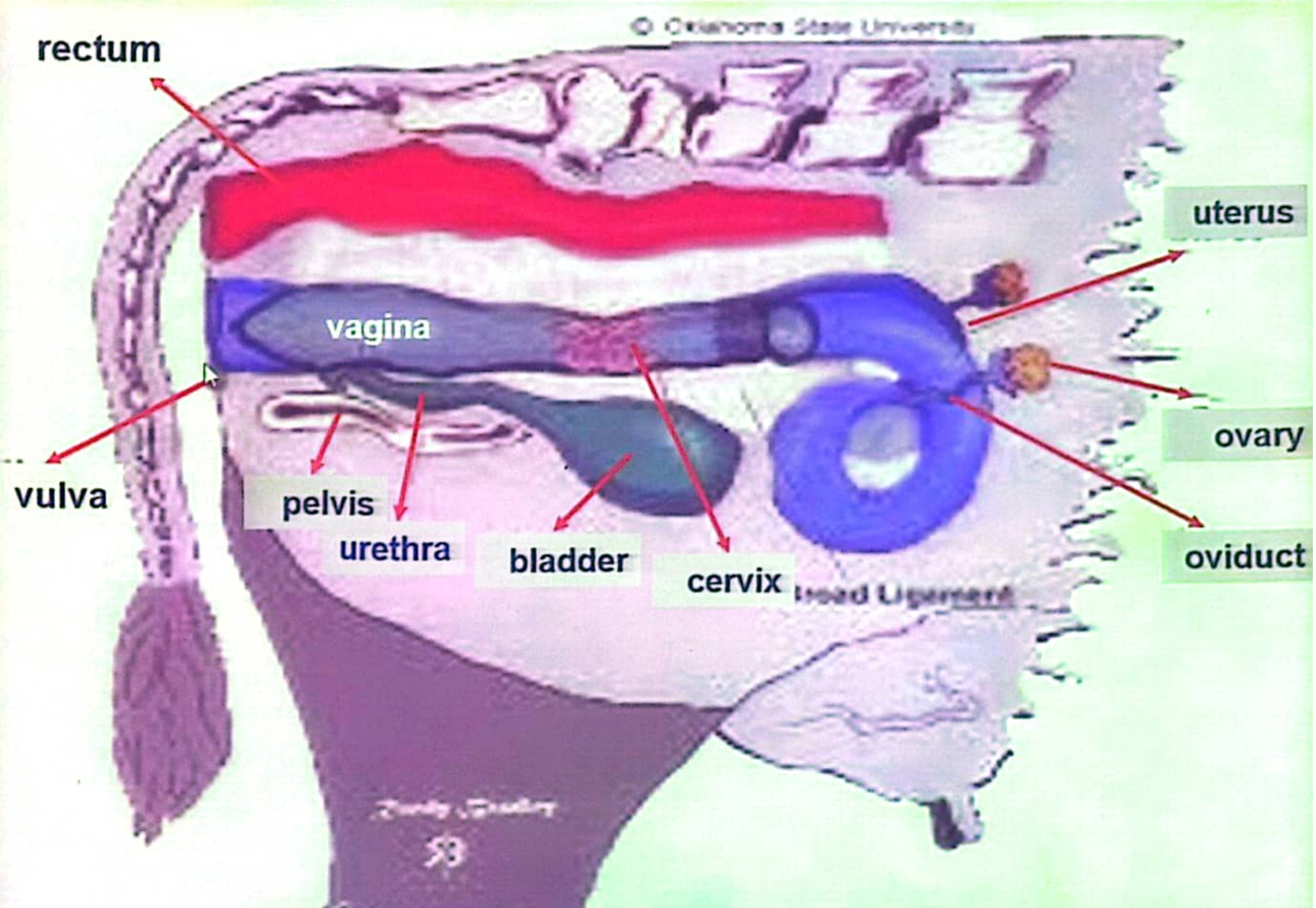
a- 2 uterine horns. b- Body of uterus.

c- Uterine cervix =(Internal cervical orifice, cervical canal, portio-vaginalis uteri and external cervical orifice).

4. Female copulatory organ include the following parts:

- Vagina (proper and vestibule). - Vulva and clitoris.
-Accessory genital glands of female (Great vestibular glands and Small vestibular glands).

5- female urethra



Female genital tract of cow

Ovary

Def:

It is the essential reproductive glands of the female; it has both endocrine (Estrogen and progesterone hormones) and cytogenic (ova) functions.

NB: It can be palpated per rectum in mare , cow, buffalo and she camel

Position:

The position of the ovaries in the different domestic animals may be in:

Sublumbar region:

- Bitch. ,Cat. ,Mare. ,She Camel.

Pelvic inlet:

-cow ,Buffalo , Sheep and goat.

Intra-pelvic: - Heifers.

Structure of the ovary=morphology

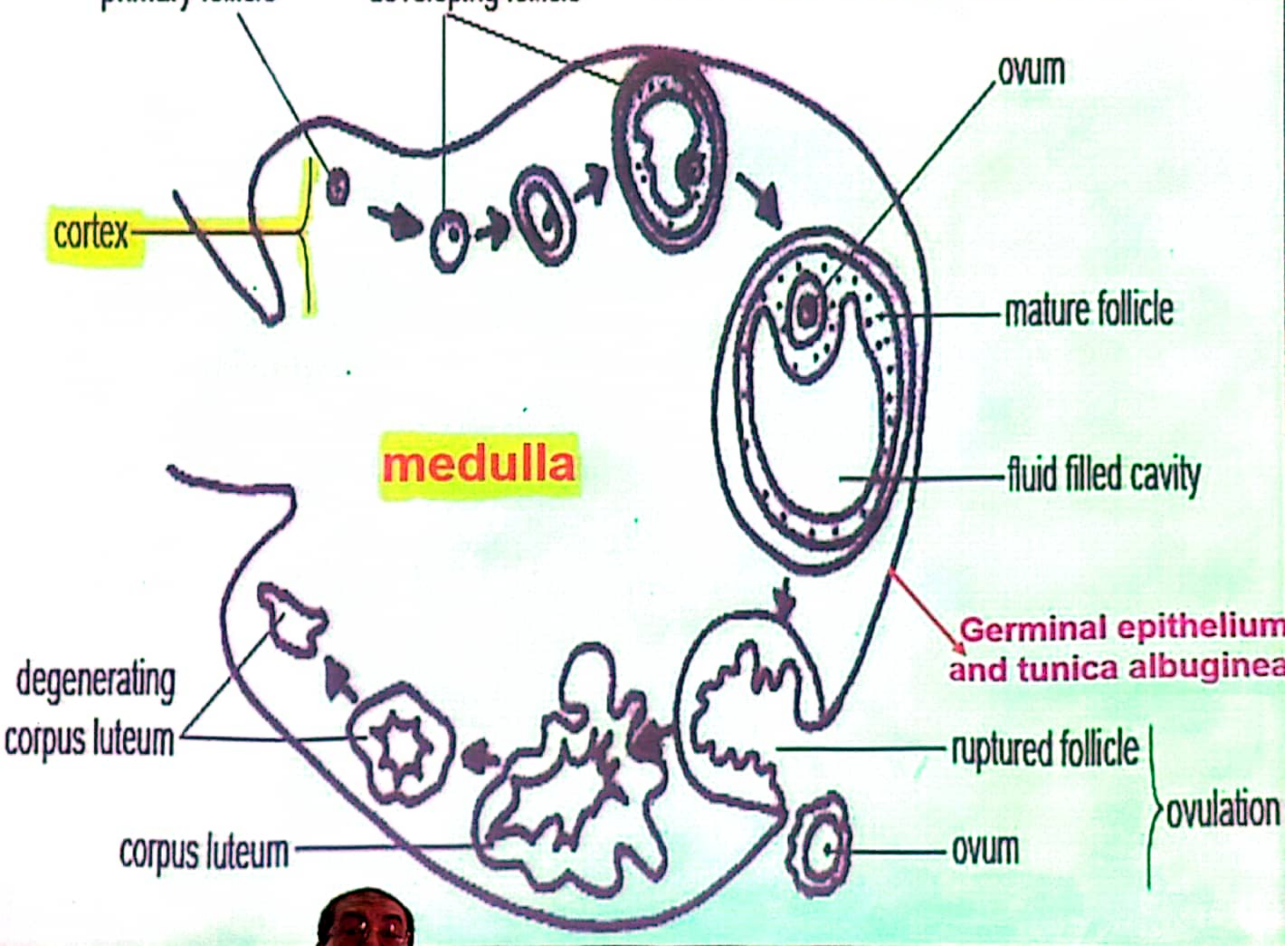
*generally, in all animals except mare it consists of outer cortex and inner medulla.

*in mare it contain outer medulla and inner cortex, so it has on its medial border an ovulation fossa.

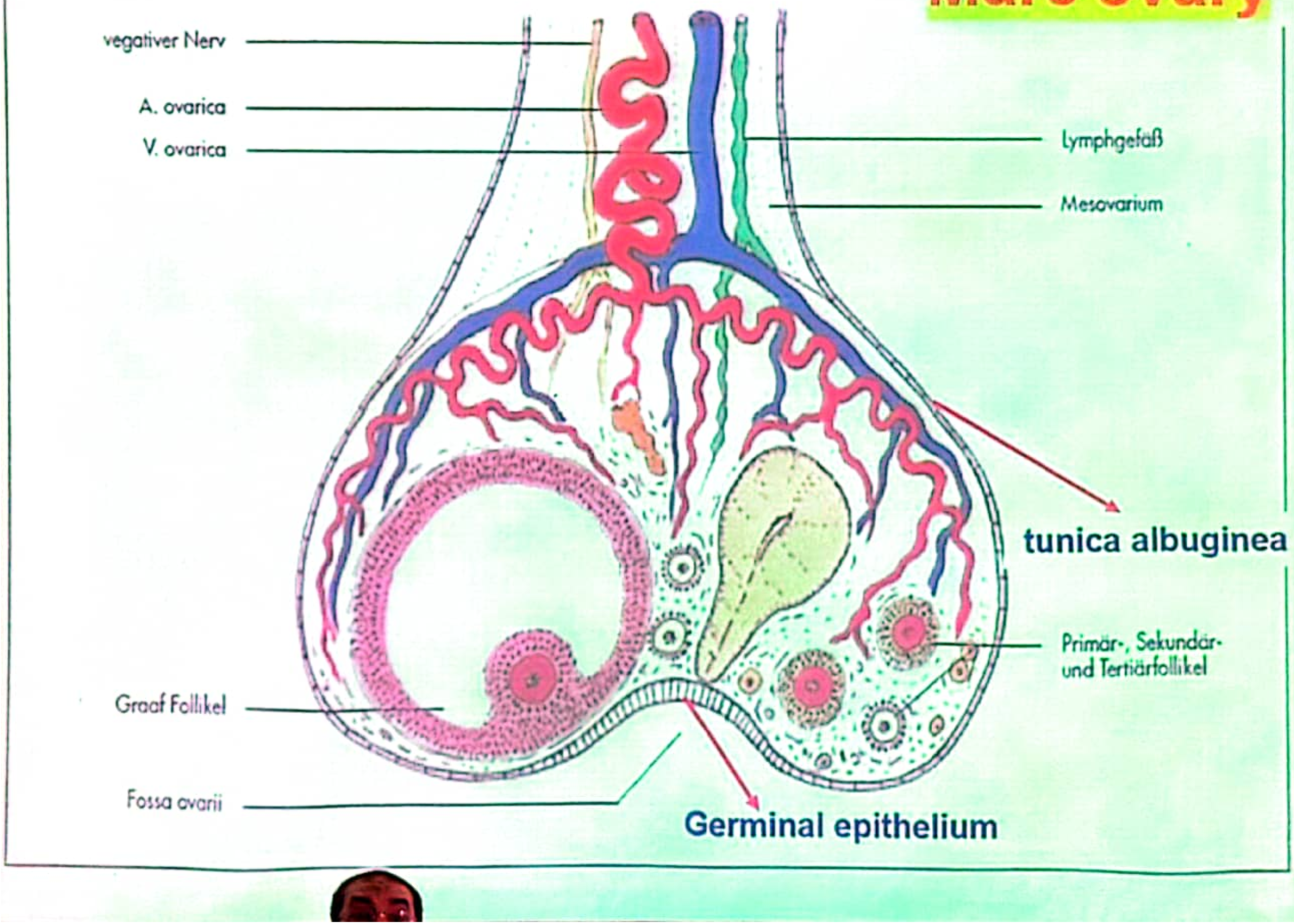
Cortex= layer contain follicles

medulla = layer contain blood ,nerve and lymph supply

Ovary of all animals except Mare



Mare ovary



vegativer Nerv
A. ovarica
V. ovarica
Lymphgefäß
Mesovarium
tunica albuginea
Primär-, Sekundär- und Tertiärfollikel
Graaf Follikel
Fossa ovarii
Germinal epithelium

Structure of the ovary

it consists of

1- germinal epithelium

2-tunica albuginea

3-cortex

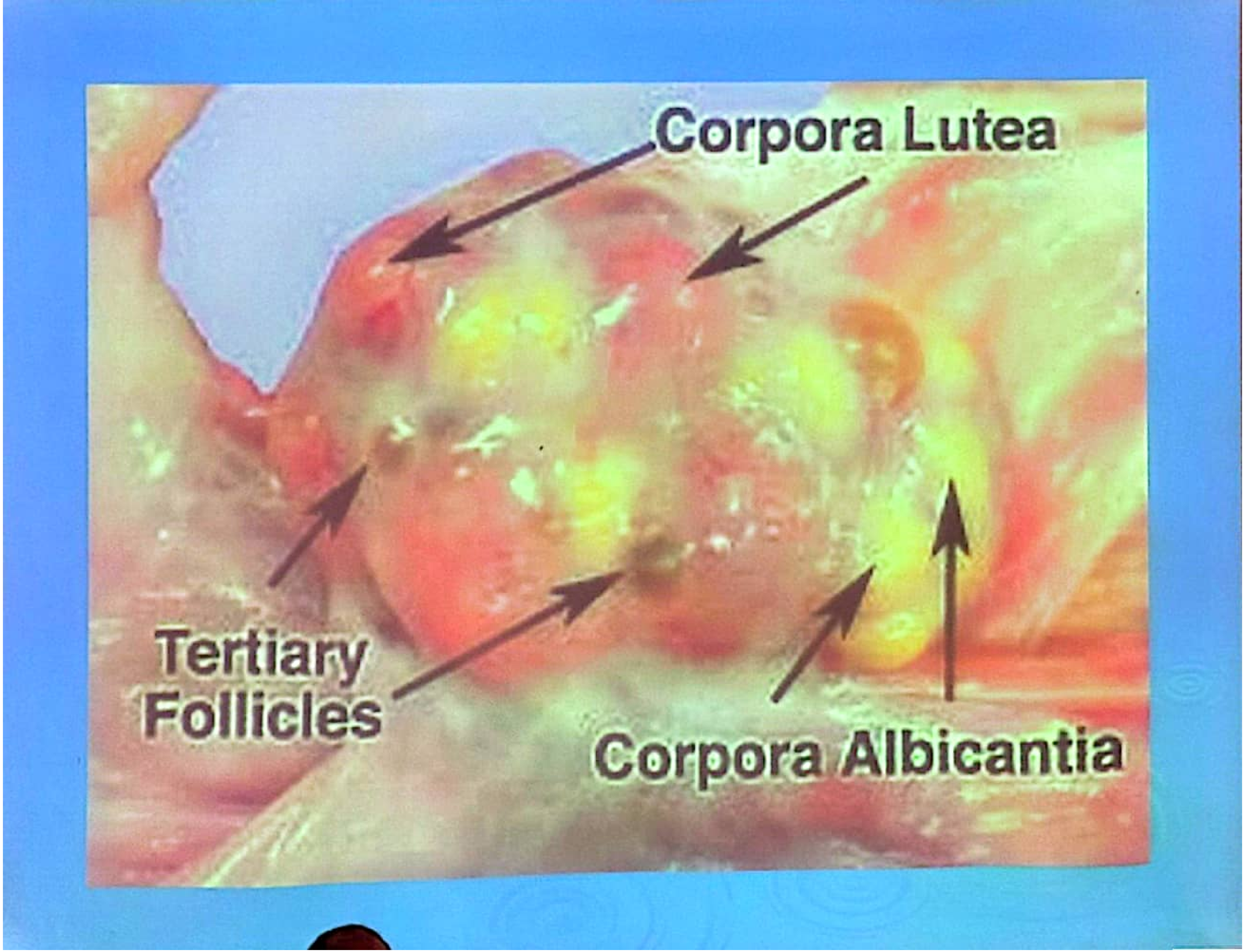
4-medulla

In mare, the ovulation fossa present as a depression on the middle of the free border of the ovary Only the **ovulation Fossa** is covered by **germinal epithelium** The cortex is situated centrally surrounding the ovulation fossa while the medulla is situated superficially surrounding the cortex except at the ovulation fossa ,So **in mare** the **surface of the ovary is smooth** as the graafian follicles and the corpus luteum do not project from it.

NB :-

In other animals which have no ovulation fossa, all the surface is covered by germinal epithelium except the attached border which is covered by serosa (Mesovarium).

The medulla is centrally placed, and the cortex is superficially, In such animals , the surface of the ovary is uneven and lobulated due to the projection of the graafian follicles and corpus luteum.


Corpora Lutea
Tertiary Follicles
Corpora Albicantia

Shape of the ovary in domestic animals may be: -

-Oval: Bitch - Cat. - Cow. - Sheep and goat.

-Ovoid (spherical) in buffalo.

- Rounded in she-camel.

-Bean-shaped in mare.

Surface of ovary: -

-Smooth: All animal except she-camel

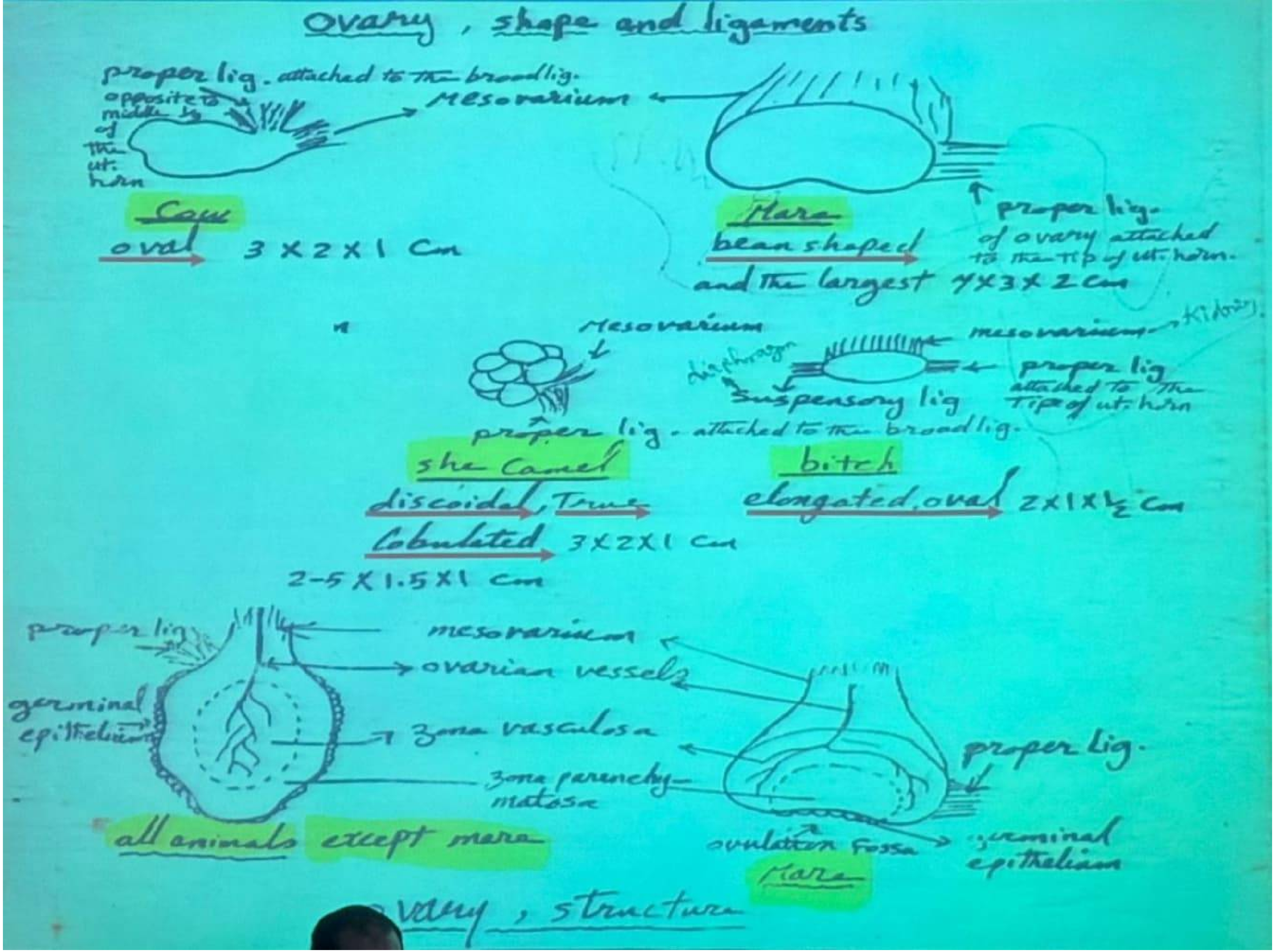
- Lobulated She-camel(true lobulated)

Morphology: Each ovary has:-

2 surfaces: - Medial. And Lateral.

2 borders: Mesoovarian border (attached border) and Free border.

2 extremities: Tubal and Uterine ends.


Ovary, shape and ligaments
proper lig. attached to the broad lig.
opposite to middle of the ut. horn
Mesovarium
Cow
oval
3 X 2 X 1 Cm
Mare
bean shaped
and the largest 7 X 3 X 2 Cm
proper lig. of ovary attached to the tip of ut. horn.
Mesovarium
proper lig. attached to the broad lig.
she Camel
discoidal, True
lobulated
3 X 2 X 1 Cm
2-5 X 1.5 X 1 Cm
mesovarium
kidney
diaphragm
suspensory lig
proper lig attached to the tip of ut. horn
bitch
elongated, oval
2 X 1 X ½ Cm
proper lig.
mesovarium
ovarian vessels
germinal epithelium
zona vasculosa
zona parenchymatosa
proper lig.
all animals except mare
ovulation fossa
germinal epithelium
Mare
ovary, structure

Ovary, shape and ligaments

Cow

oval → 3 X 2 X 1 Cm

Mare

bean shaped →
and the largest 7 X 3 X 2 Cm

she Camel

discoidal, True → lobulated → 3 X 2 X 1 Cm
2-5 X 1.5 X 1 Cm

bitch

elongated, oval → 2 X 1 X ½ Cm

mesovarium
ovarian vessels
Zona vasculosa
Zona parenchymatosa

all animals except mare

Mare

Ovary, structure

Fixation of the ovary=ligaments

1- mesovary

it is part of broad ligament attached between ovary and sublumbar region (in mare. she-camel and bitch)
or to the lateral pelvic wall and to the flank region (in cow & ewe) ,
it contain blood vessels ,nerves and lymph through ovarian hilus

2-proper ligament of the ovary

connect the ovary and the cranial part of uterine horn

3-suspensory ligament of the ovary

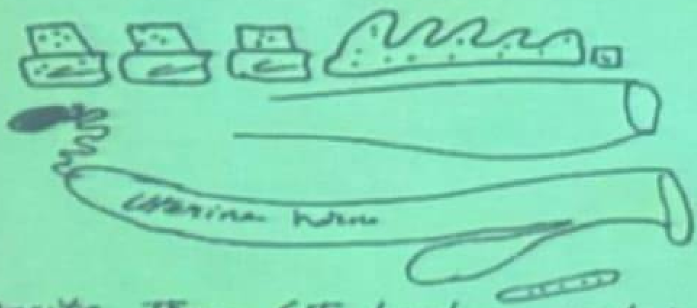
connects between ovary and diaphragm in case of dog only

Ovarian bursa

it is serous sac formed by proper ligament of ovary ,mesovary and mesosalpinx
The ovary either lies partially as in case of mare , cow ,buffalo and cat
or completely as in bitch and she-camel.

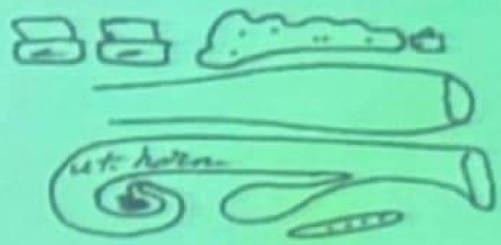
Position of the ovary

In the sublumbar region



opposite the 6th lumbar vert. in camel
4th & 5th " " " mare
3rd & 4th " " " bitch

at the pelvic inlet



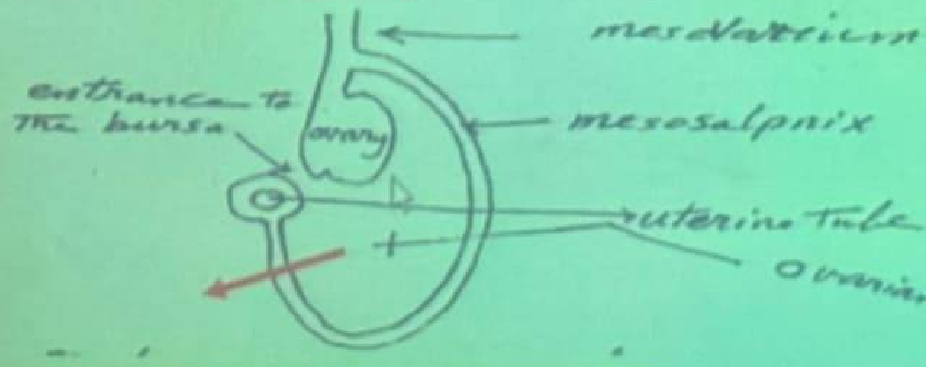
- at the middle of its lateral margin in the coils of the uterine horn as in cow

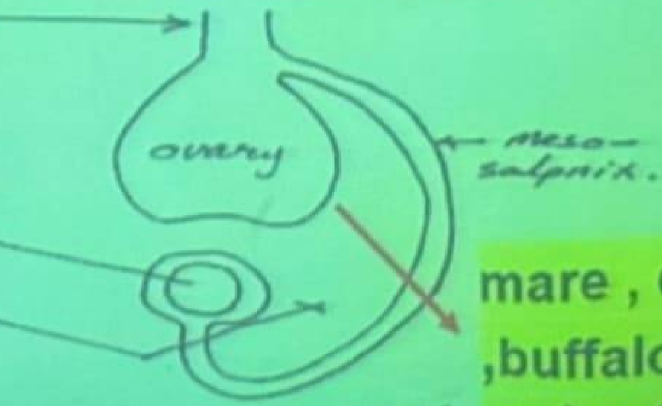
Ovarian bursa

Completely conceal the ovary

partially cover the ovary

mesovarium





bitch and sh[illegible]amel

mare , cow ,buffalo and cat

Position of the ovary

In the sublumbar region

opposite the 6th lumbar vert. in camel
4th & 5th " " " Mare
3rd & 4th " " " bitch

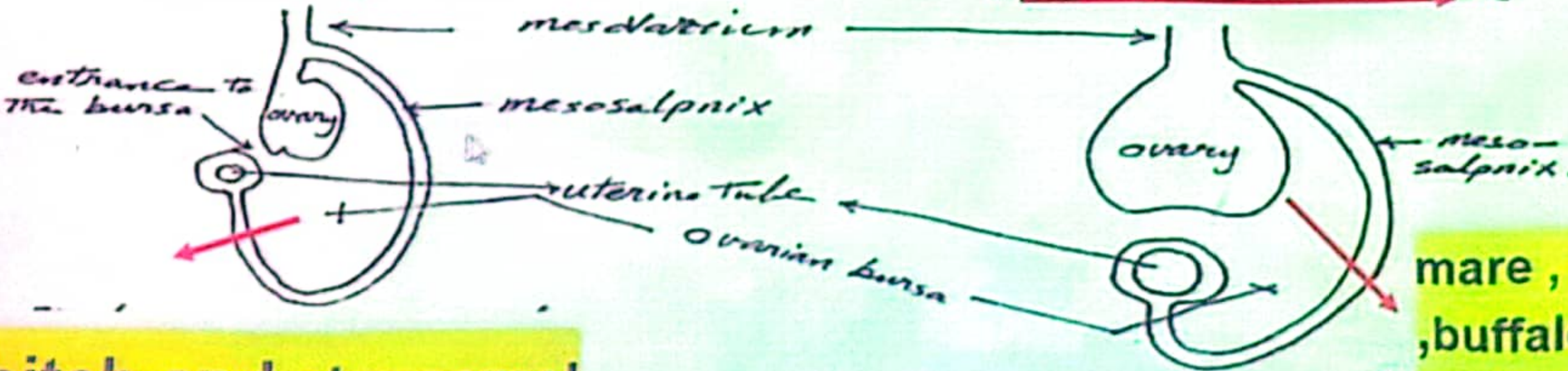
at the pelvic inlet

- at the middle of its lateral margin in the coils of the uterine horn as in Cow

Ovarian bursa

Completely conceal the ovary

partially cover the ovary

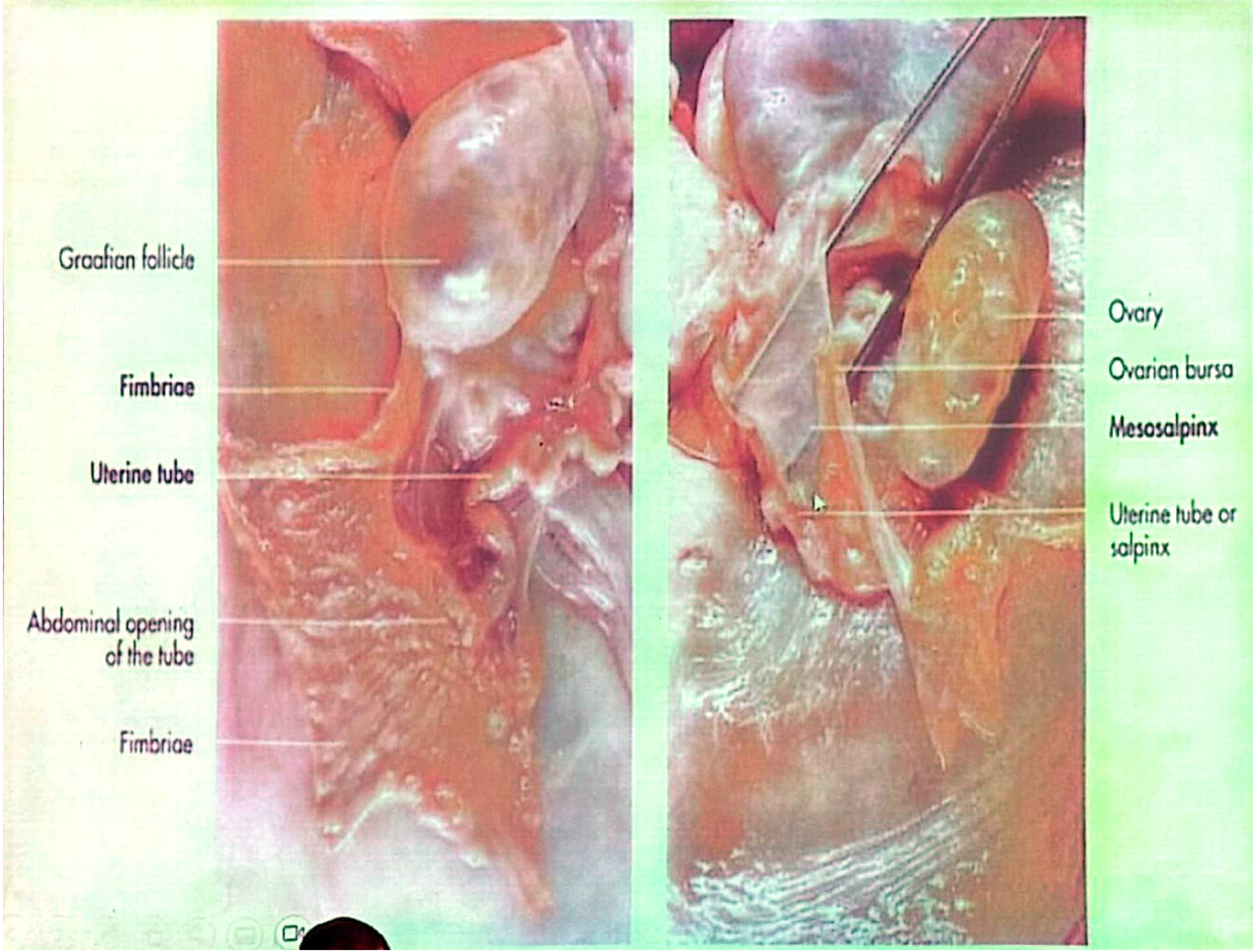


bitch and she camel.

mare, cow ,buffalo and cat

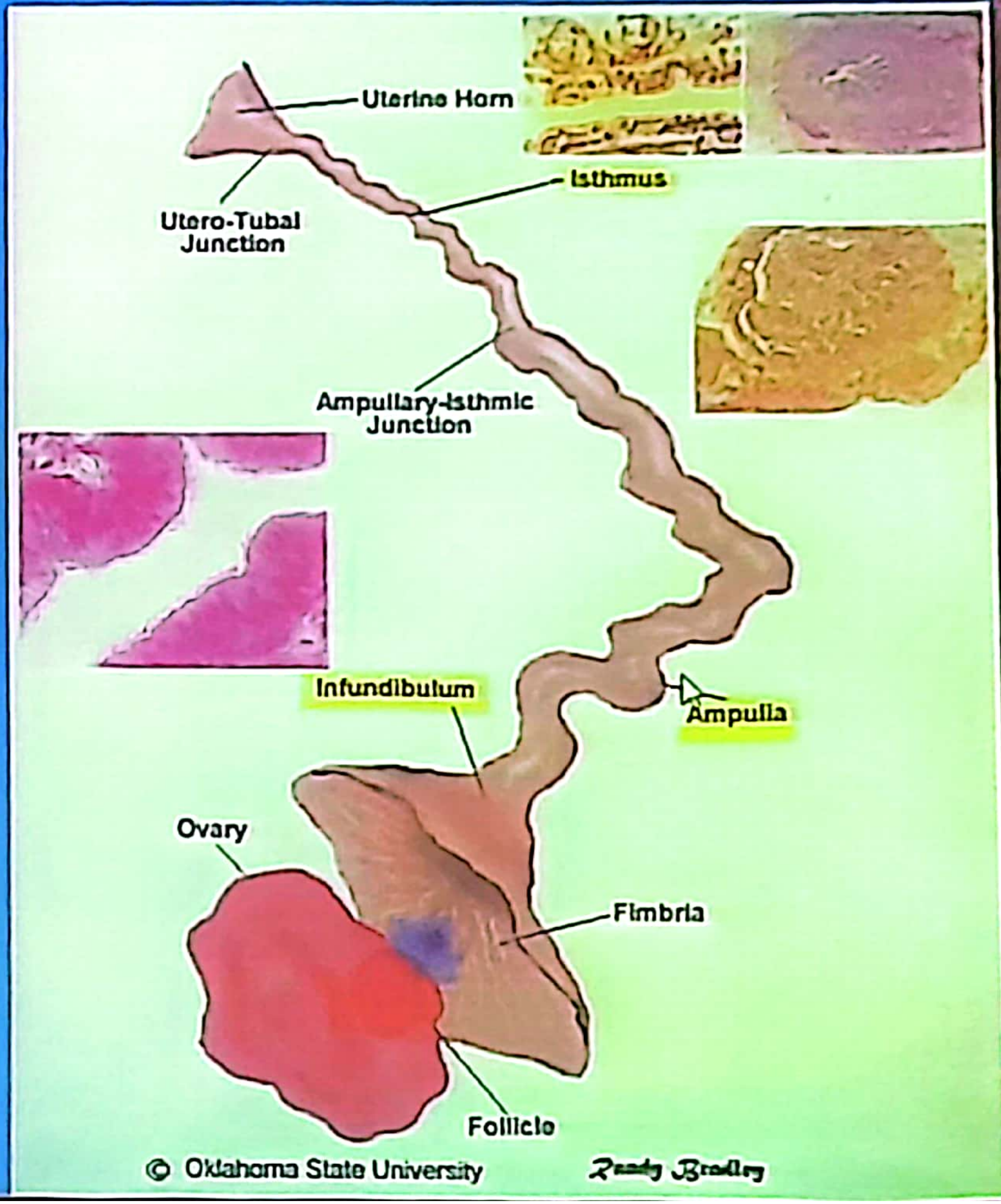
Comparative ovary

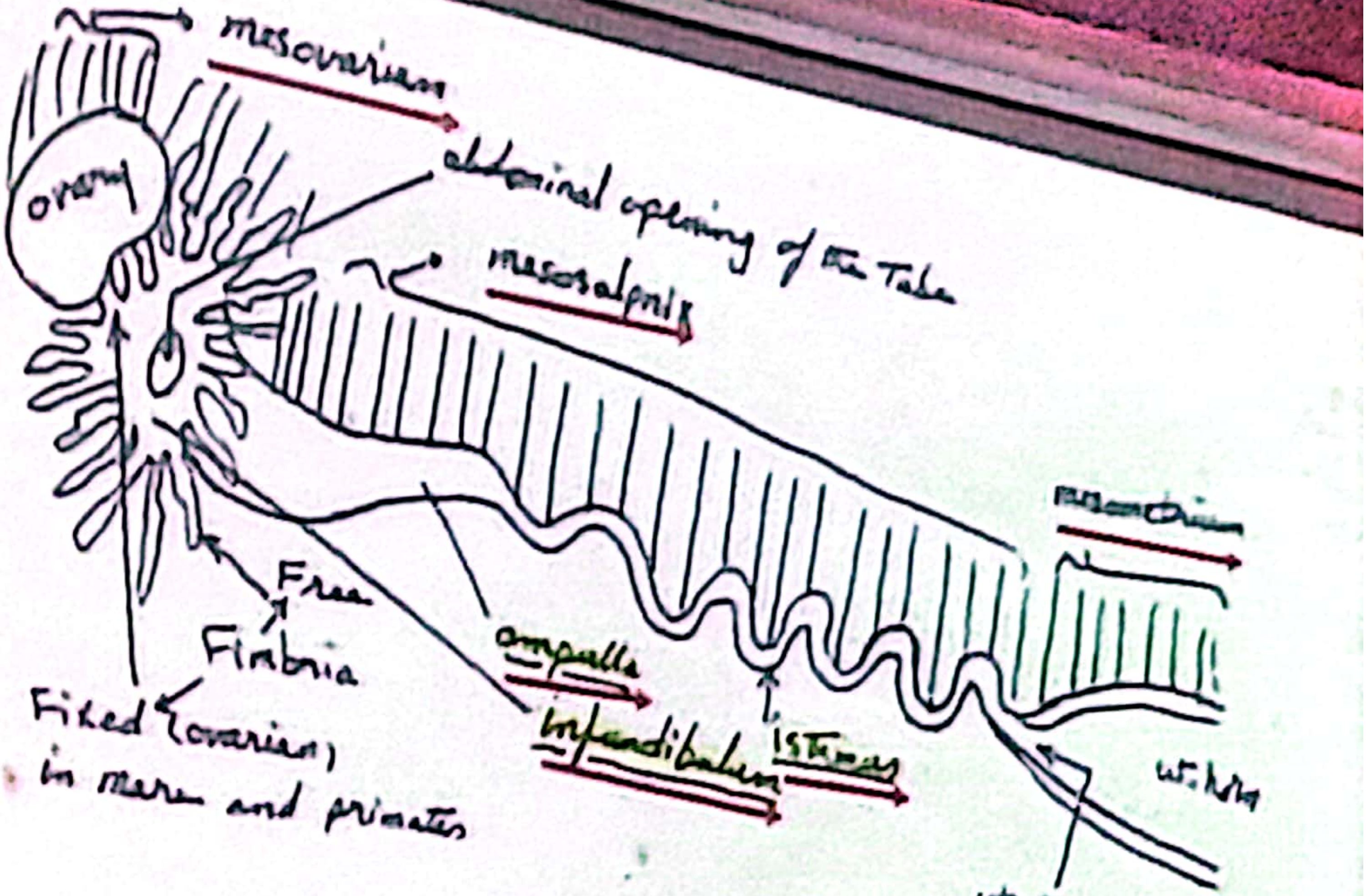
Points of Comparative	Mare	Cow	She camel	Bitch
Shape	Bean	Oval	Circular	Oval
External surface	Smooth	Smooth	Lobulated	Smooth
Position	Sublumbar	Pelvic inlet	Sublumbar	Sublumbar
Ovulation fossa	present	Absent	Absent	Absent
Structure	Cortex inside Medulla outside	Cortex outside Medulla inside	As cow	As cow
Ovarian bursa	Partially situated	Partially situated	Completely situated	Completely situated
Ligaments	Mesovarium & proper	Mesovarium & proper	Mesovarium & proper ligs	Mesovarium . proper& Suspensory


Graafian follicle
Fimbriae
Uterine tube
Abdominal opening of the tube
Fimbriae
Ovary
Ovarian bursa
Mesosalpinx
Uterine tube or salpinx

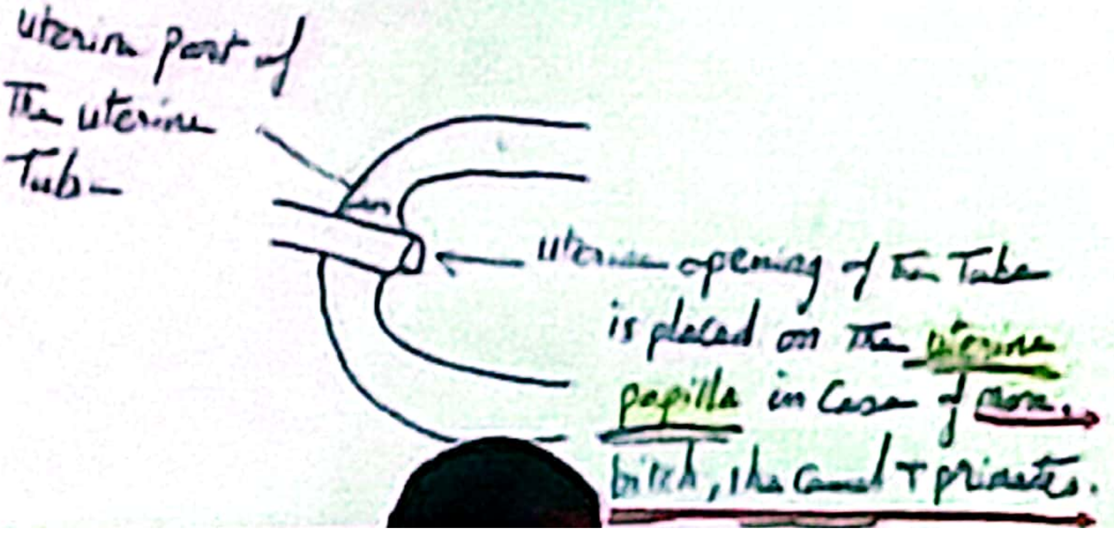
Oviduct = uterine tube=Fallopian tube

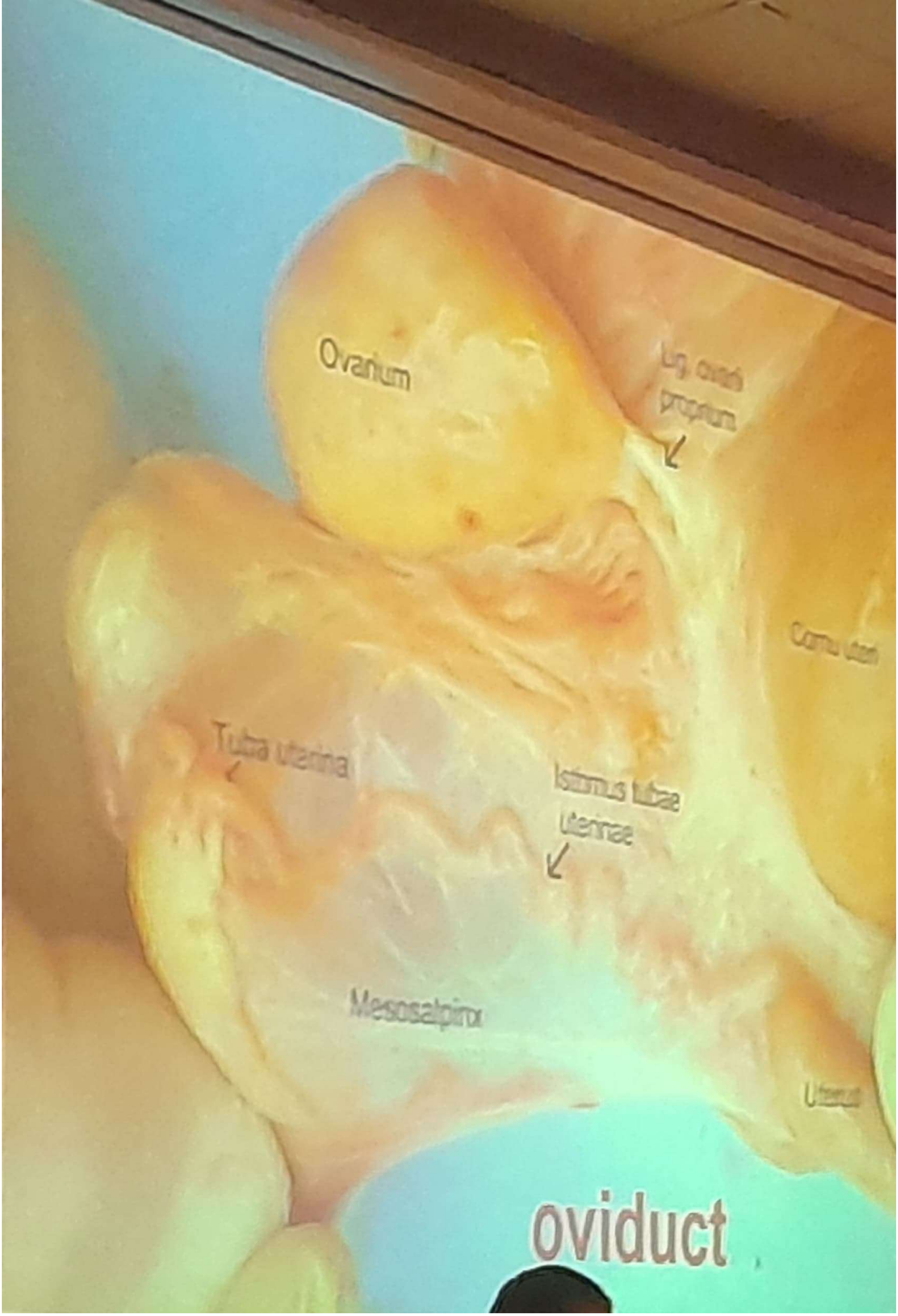
- It consists of three parts:-
- 1- infundibulum = funnel shape with fimbriae.
- 2-ampulla = dilated part for fertilization.
- 3-isthmus = flexuous, it is ended at uterine horn
- (gradually in cow , ewe) or with (papilla in mare ,bitch and she –camel)





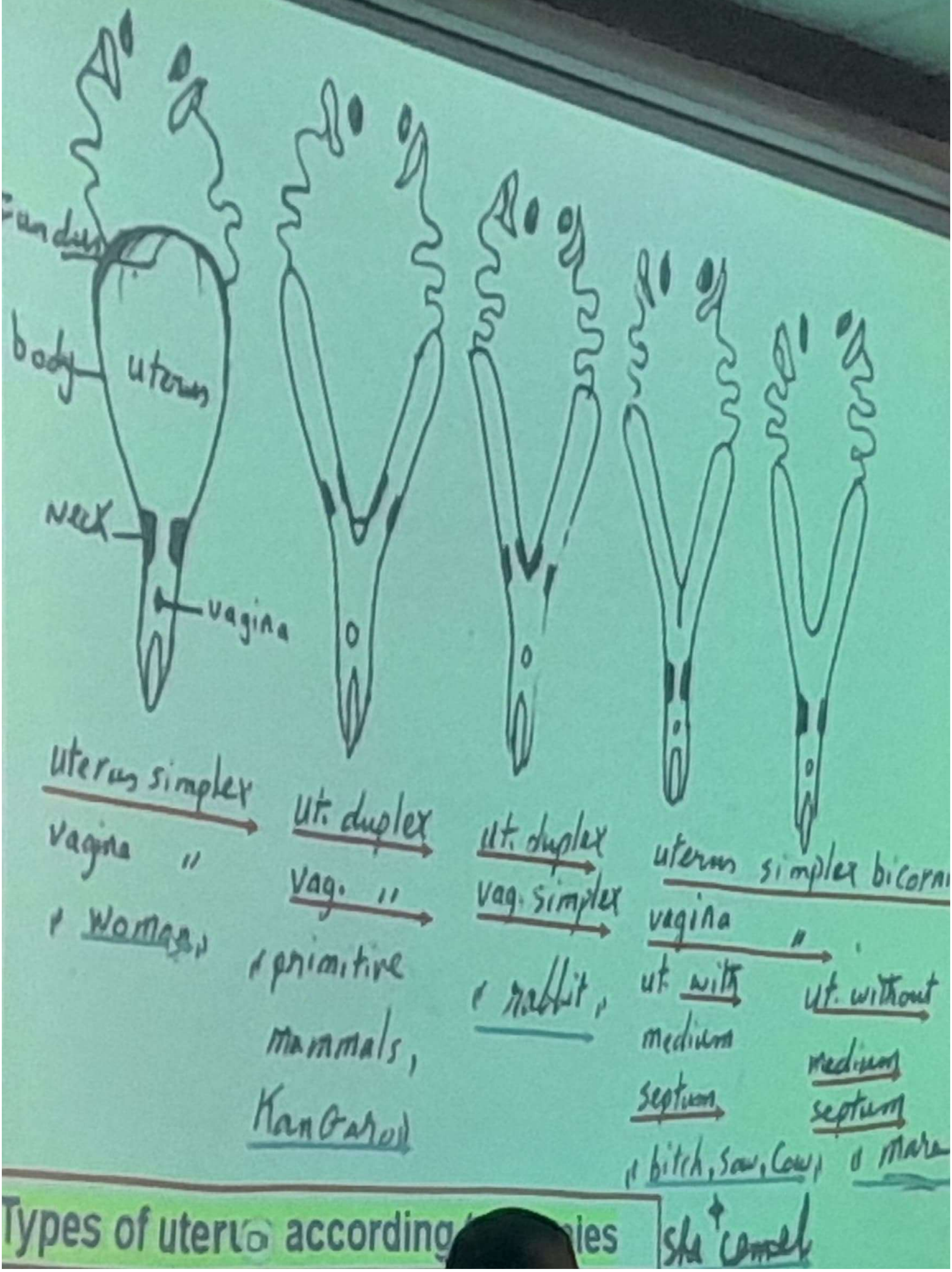
Uterine tube




Ovarium
Lig. ovarii proprium
Cornu uteri
Tuba uterina
Isthmus tubae uterinae
Mesosalpinx
Uterus
oviduct

The abdominal ostium of infundibulum

leads to the **ampulla** ,where fertilization usually takes place. The oocyte remains in the ampulla for a few days before it is transported to the **apex of the horn of the uterus** through the narrower, more convoluted distal part of the tube, **the isthmus**. The uterine tube opens into the uterine horn through the **uterine ostium** and marks the site of the uterotubal junction. The junction is gradual **in ruminants** but shows an **abrupt junction in the horse she camel and in carnivores**, in which the uterine ostium is located on top of a papilla, thus forming a barrier against ascending infections.


fundus
body
uterus
neck
vagina
uterus simplex
vagina "
e woman
ut. duplex
vag. "
e primitive mammals, Kangaroo
ut. duplex
vag. simplex
e rabbit
uterus simplex bicorn
vagina "
ut. with medium septum
e bitch, Sow, Cow
ut. without medium septum
e mare
she camel
Types of uteru according
ies

uterus

* it is a hollow muscular organ with thick muscular wall (myometrium)
* It receive s the fertilized ova on its mucous coat =(endometrium) where the fertilized ova implantation takes place embryo till the fetus birth.
* the outer surface or serous coat of uterus is covered by visceral peritoneum (epimetrium)
* the perimetrium is the line of attachment of mesometrium (=uterine part of broad ligament)with the epimetrium

NB in case of bitch there is round ligament of uterus as fixation of uterus connect between uterine horn and inguinal ring

position of uterus :-

In ruminants' part within abdominal and pelvic cavity (pelvic inlet)

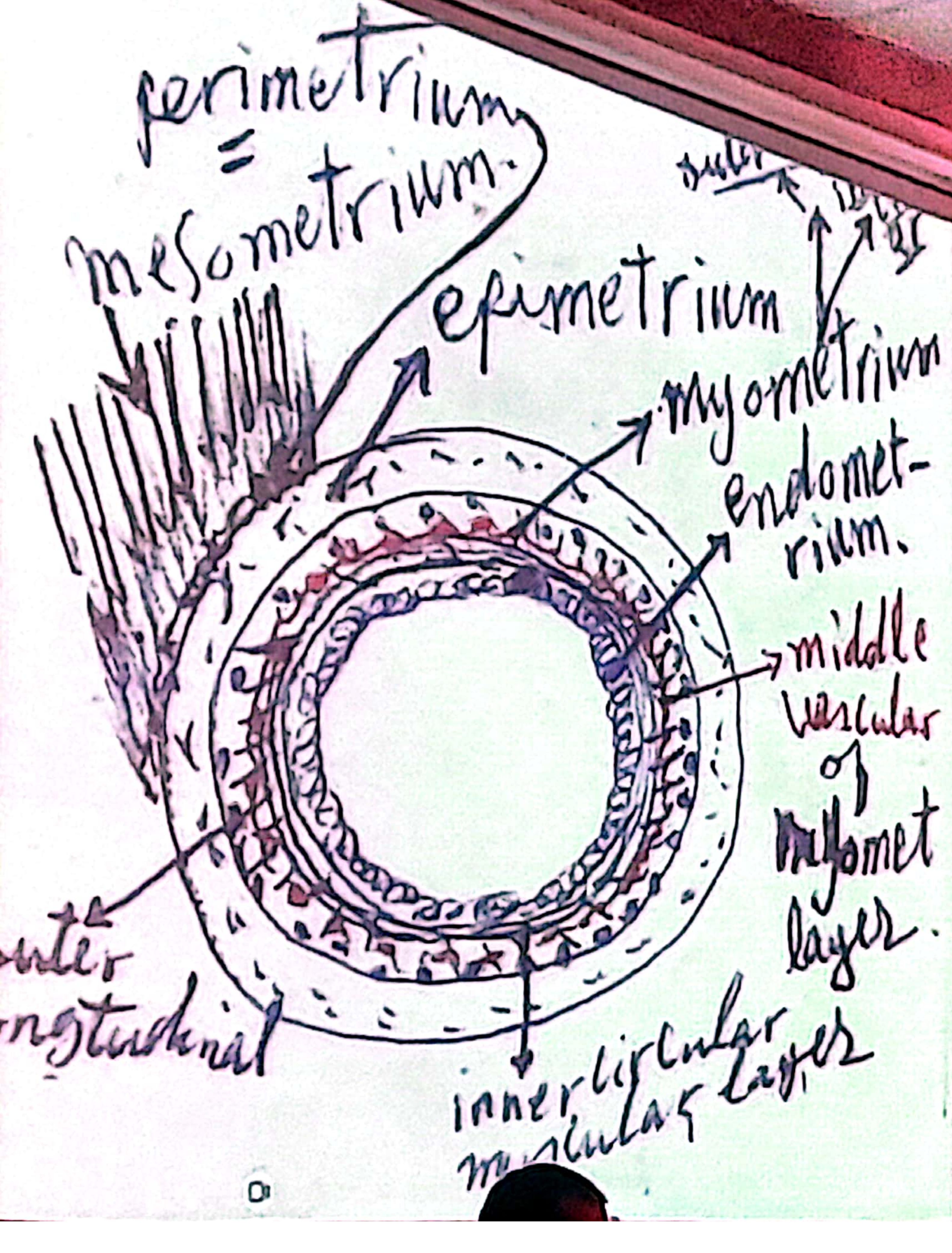
In non ruminant animals in abdominal cavity

uterine horn , coiled in cow , buffalo, sheep and goat

straight in mare and bitch T-shape in she-camel

uterine body short in ruminants and bitch

long in mare


perimetrium
=
mesometrium.
epimetrium
myometrium
endomet-
rium.
middle
vascular
of
myomet
layer.
outer
longitudinal
inner circular
muscular layer

Uterine cervix

a- internal uterine orifice leads to uterine cavity
b- external uterine orifice leads to vaginal cavity
which is found between a and b
d-Portiovaginalis uteri = that part of cervix which project into the cranial part of the vaginal cavity ...it is surrounded usually by a circular cavity known as vaginal fornix.

Vessel and nerves of the uterus

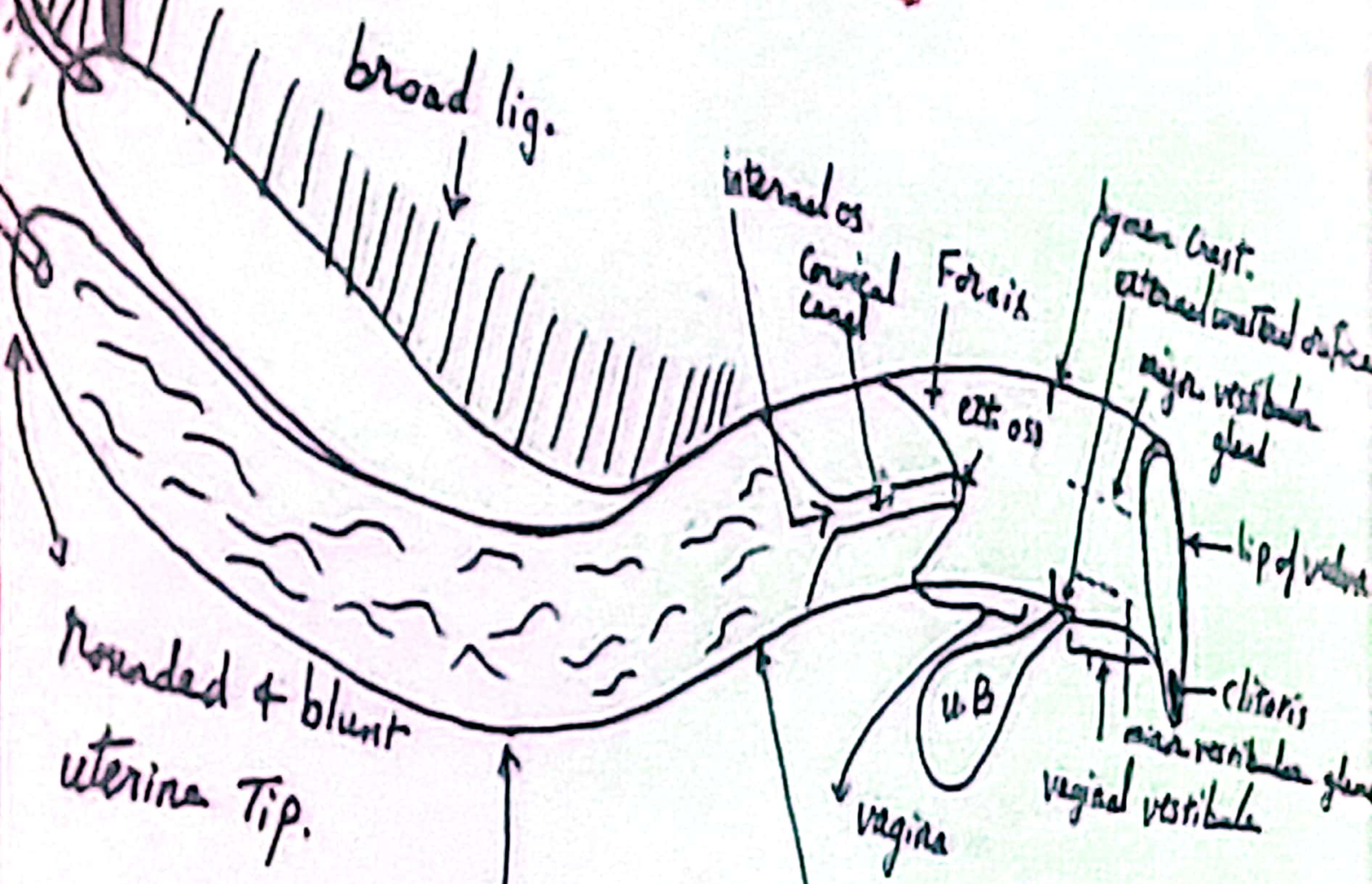
Arteries & veins ovarian and uterine V+A
Lymph vessels internal iliac and lumer lymph nodes
Nerves

Sympathetic through hypogastric plexus
Parasympathetic through the pelvic nerves

Uterine horns and body

Points of comparative	Mare	Cow	She camel	Ewe
Uterine horn shape	Straight Dorsal curvature concave, ventral convex 15-25 cm	Coiled ram horn like dorsal curvature convex, ventral concave D-v intercornual lg 35-40cm	T-Shape L.H-11cm R.H-8cm	V-Shape 17cm
Uterine body	Long 15-20 cm	Short 2-4 cm intercornual septum	Short	Short
Endometrium	Contain leaflike longitudinal folds	Contain Uterine caruncles are an oval elevations in the endometrium contains glands in its depth 70-120	smooth	smooth
Cervix	7cm only L.folds Easily dilated	Numerous L.folds crossed by 3-4 T.folds	4cm Easily dilated	Very narrow opening of cervix closed together
Fornix	Circular	Dorsal	Dorsal	Ventral
Portio-vaginalis	Clear centrally	ventral	Ventral 1cm with serrated borders	Dorsal

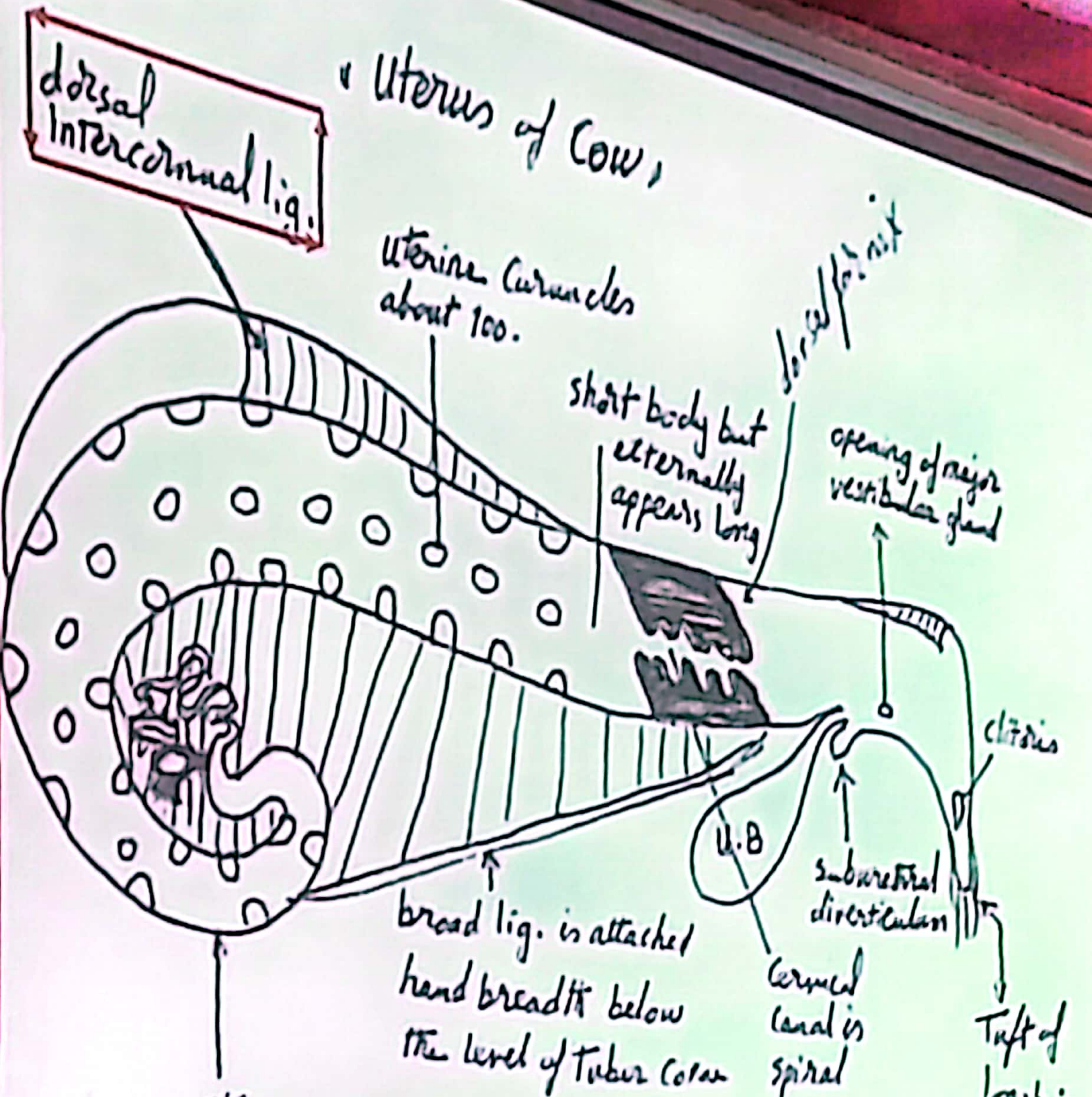
Uterus of Mare



The uterine horn nearly straight and is hanged in abdominal cavity, its endometrium is folded, has leaf-like folds

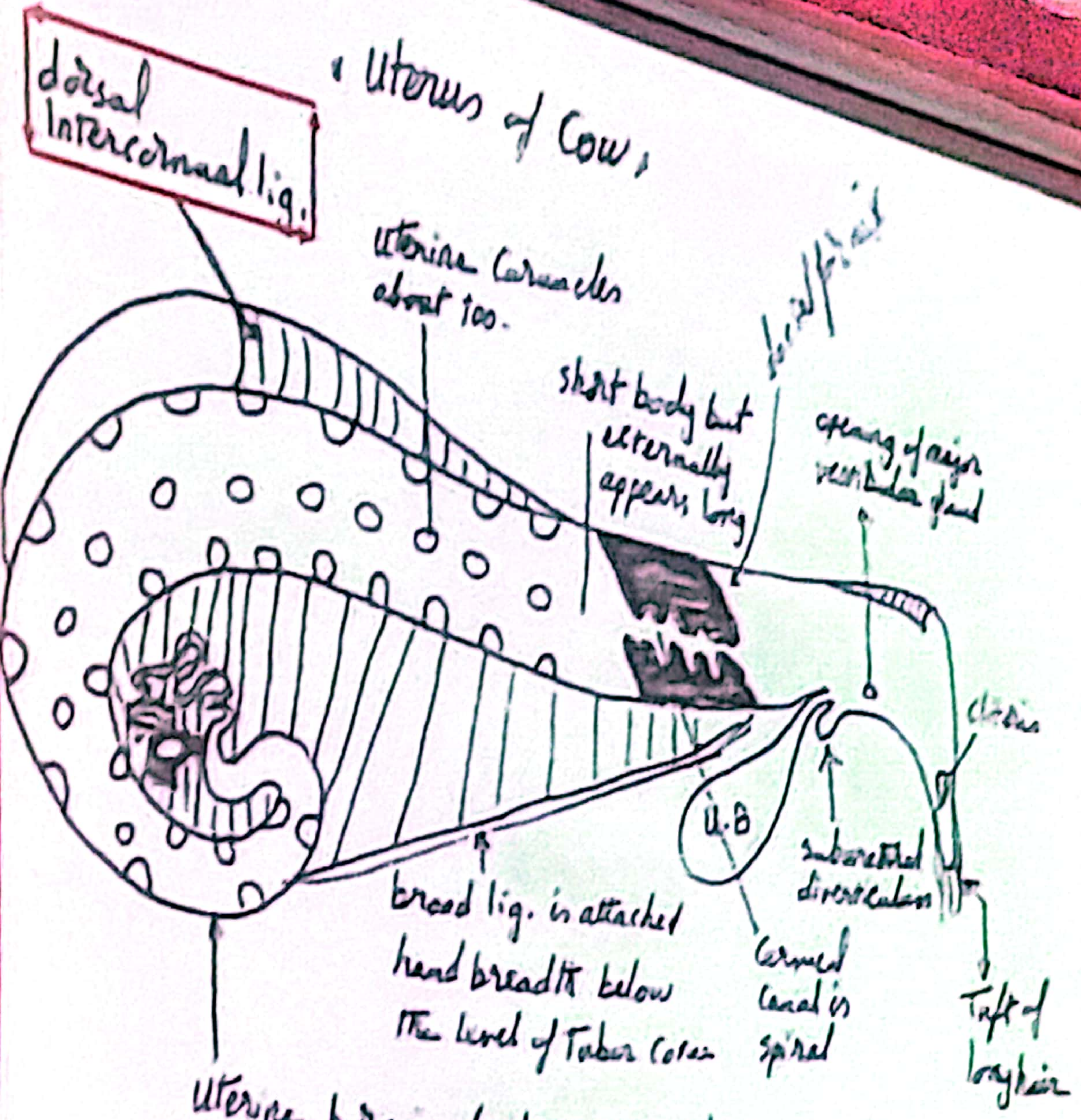
uterine body is long, 20 cm. has no Traces of median septum

Uterus of Cow,

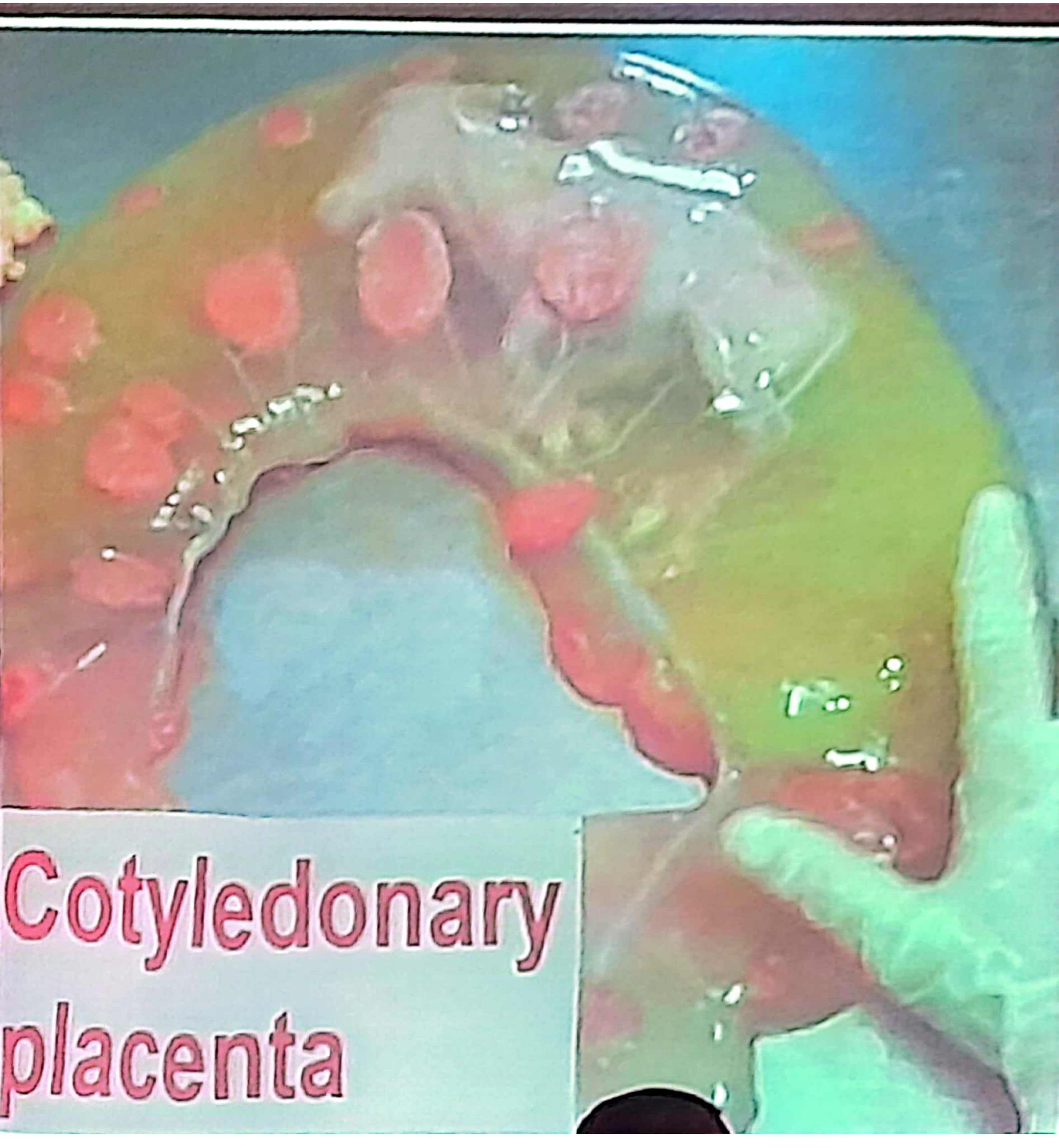


Uterine horn is about 40 cm and forms spiral Coil and Tapers gradually Toward the Tip. it opens independant into the bod ... ce of intercornual septum)

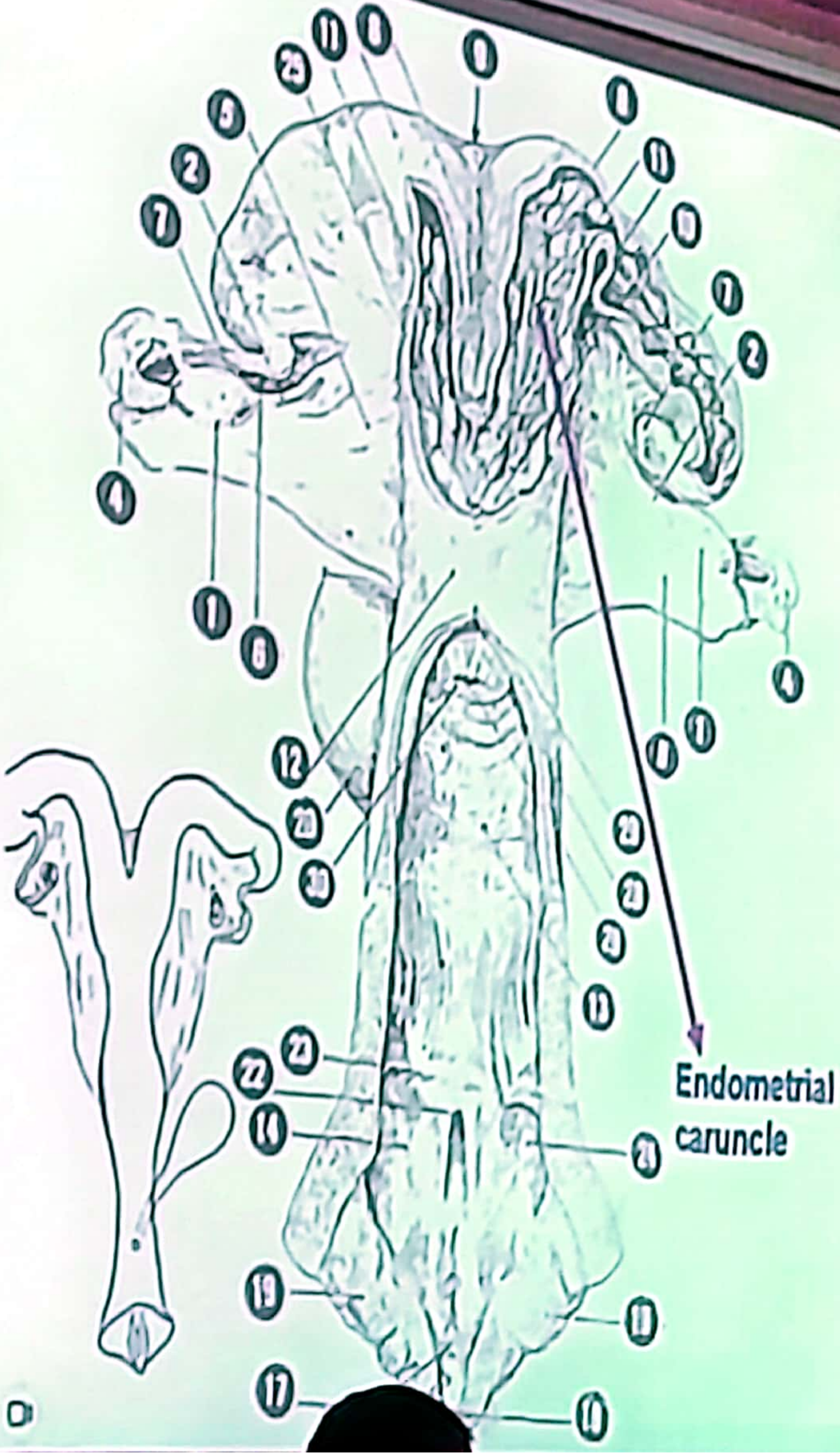
Uterus of Cow,

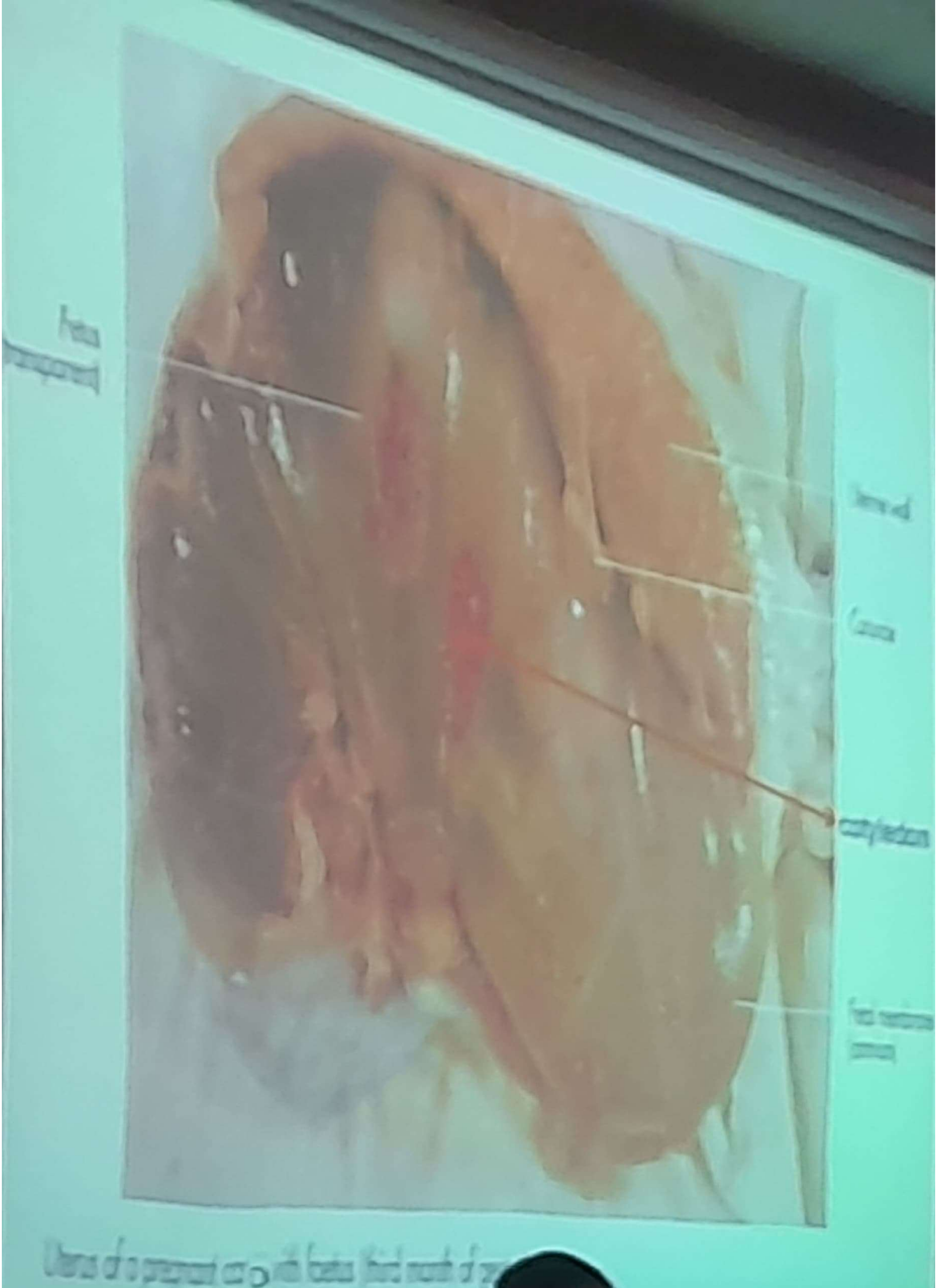


Uterine horn is about 40 cm and forms spiral coil and Tapers gradually Toward the Tip. it open independent into the [illegible] presence of intercornual septum)


Cotyledonary
placenta

Uters of cow





Uterus of a pregnant cow with foetus (third month of pr

Uters of she camel

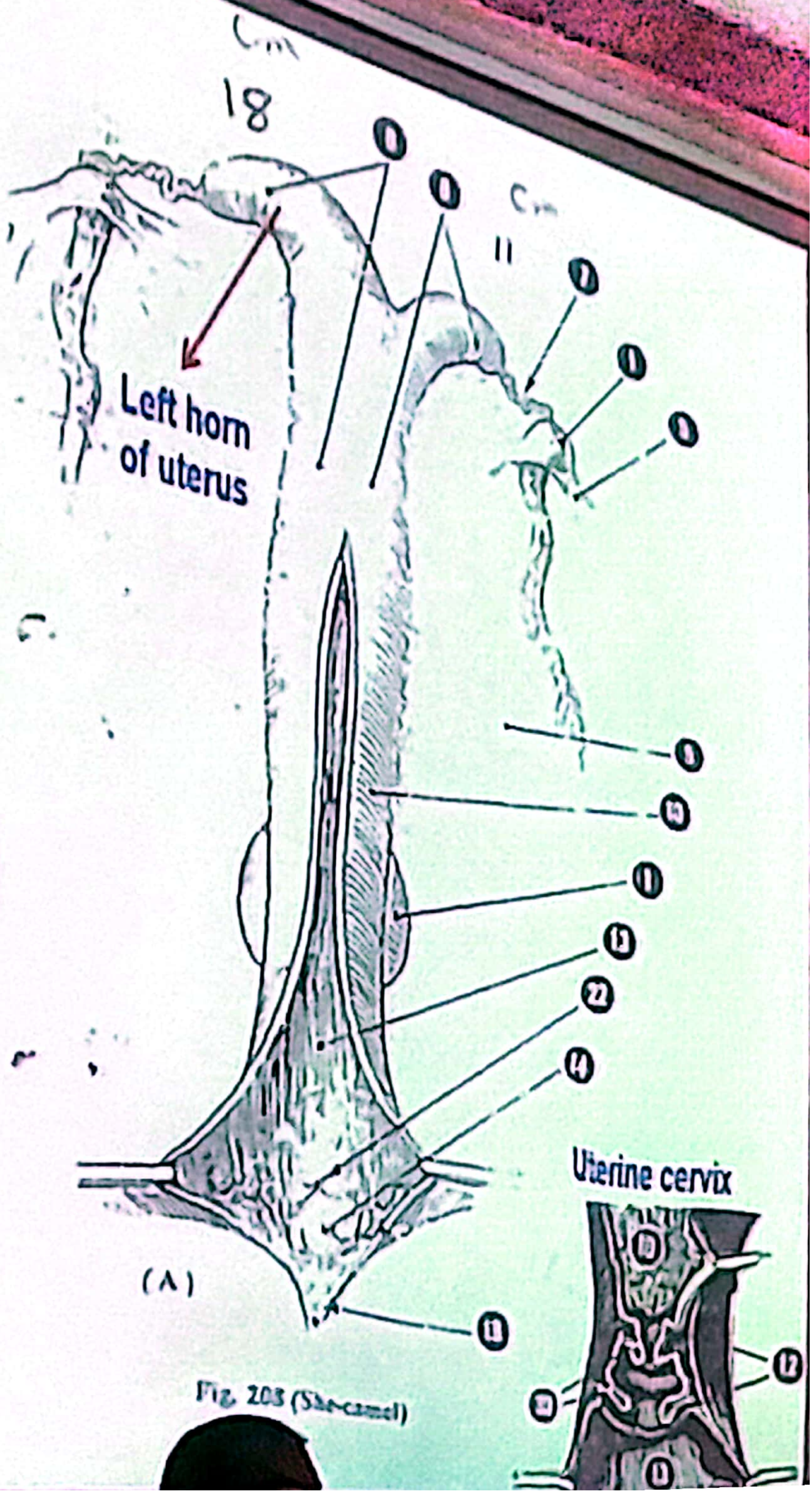
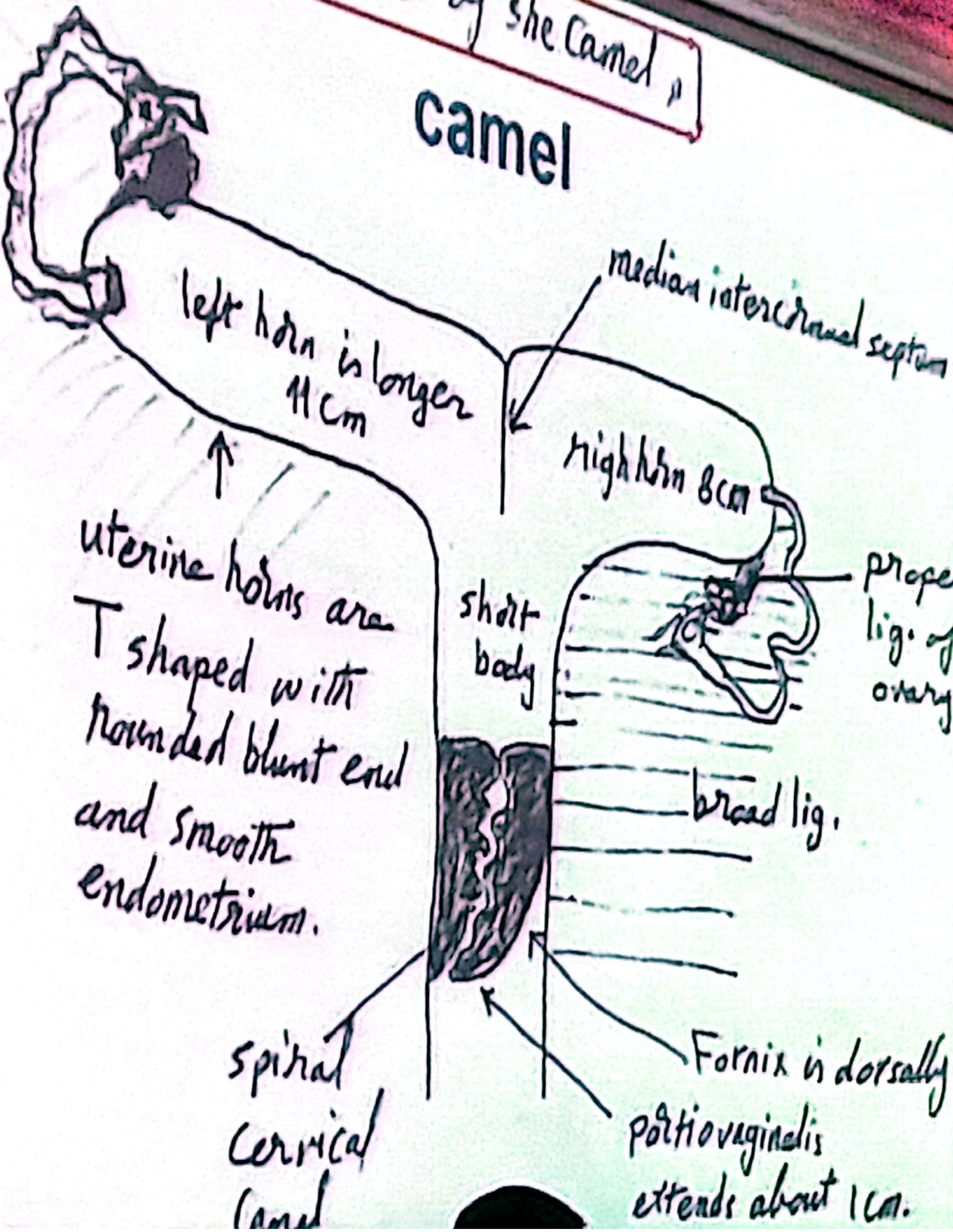


Fig. 203 (She-camel)

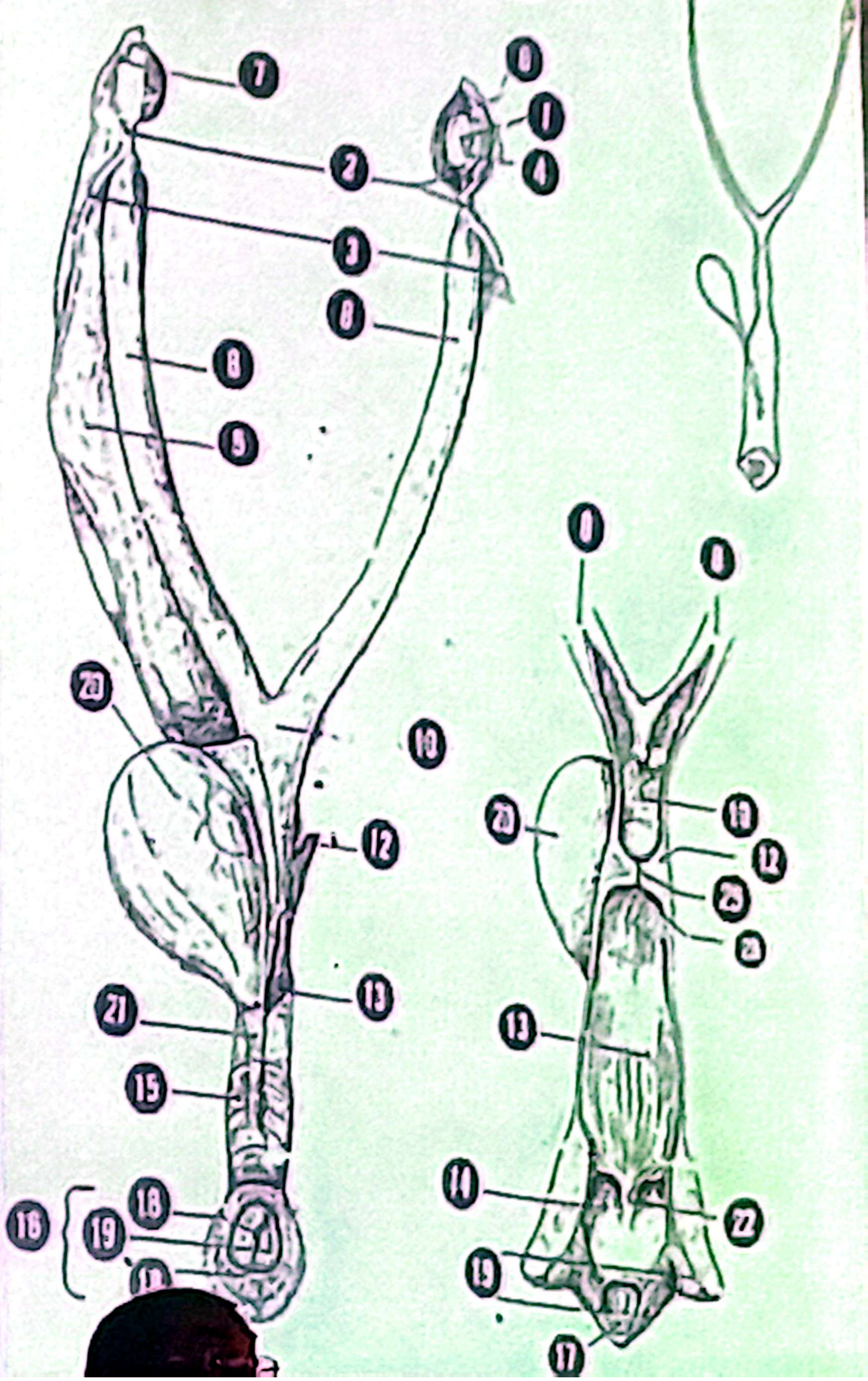
" uterus of she camel "

camel



Uterus of bitch

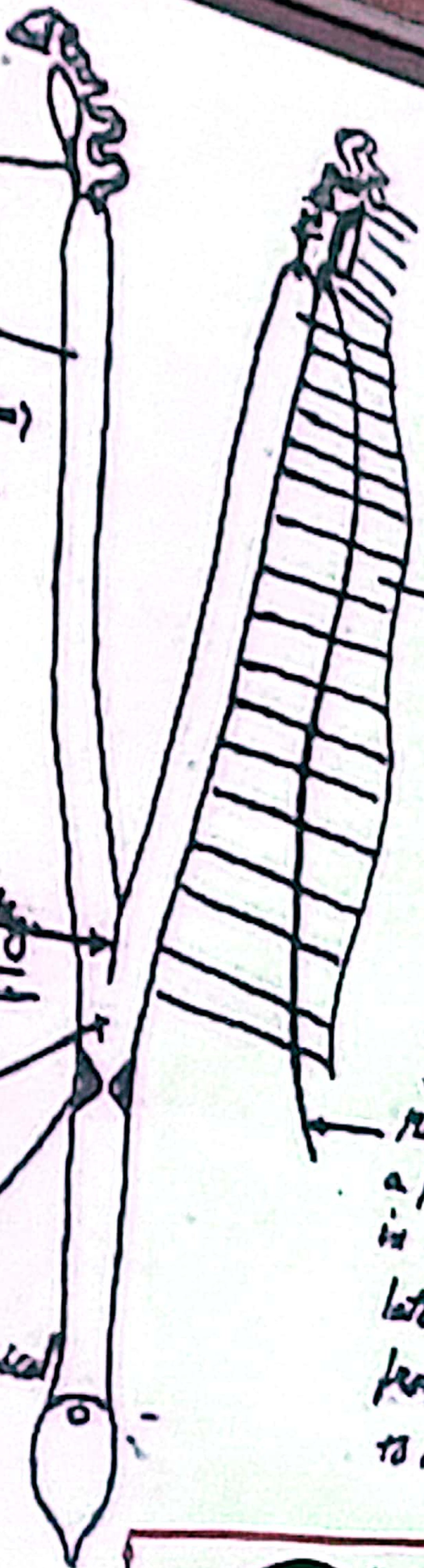
The uterus lies mainly dorsal to the small intestines. It consists of a short cervix and body from which two long slender horns diverge to reach the ovaries just caudal to the kidneys. An internal partition, which is not obvious externally, projects into the body of the uterus separating the horns. in the cow, in which the septum extends almost as far as the cervix.



bitch

proper lig.

uterine horns are long (12-15 cm) & pencile like in diameter.



mesometrium & broad lig. contains fat between its two layers.

median intercornual lig. septum divides the fundus.

body is short 1-3 cm

Cervix is thick and short no cervical canal or portio-vaginalis.

ext. os = int. os

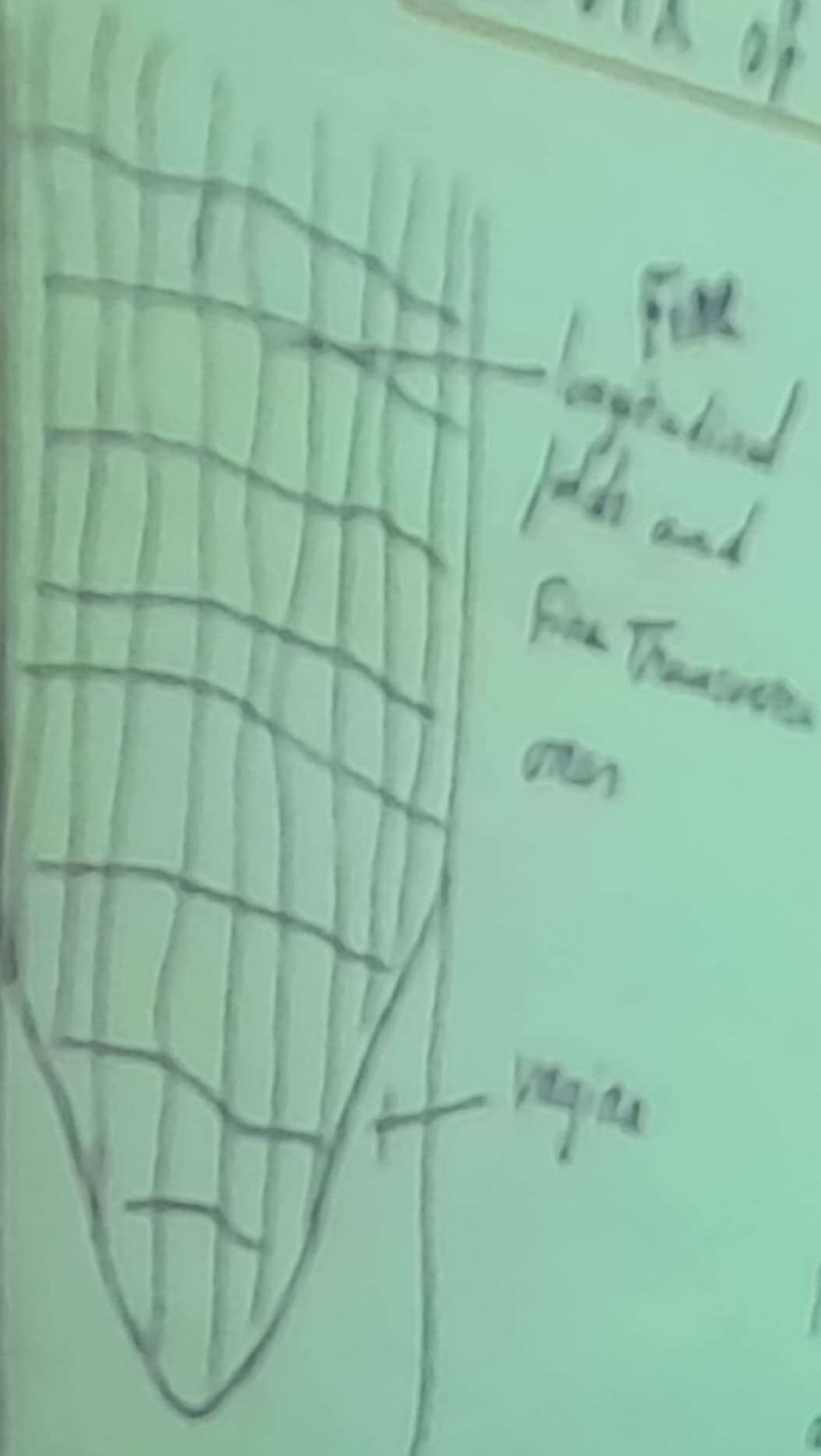
Round lig. of uterus is a fibrous cord extends in a serosal fold on the lateral surface of mesometrium from the tip of the ut. horn to the deep inguinal ring.

Bitch

2-Uterine cervix

*It is called the neck
*it is part of uterus between the uterine body and the vagina
*it has cervical canal connect between internal and external cervical orifice=cervical os (closed all times except at birth and estrous)
*some species its cervix is projected into vagina by portiovaginalis = vaginal portion of cervix and surrounded by vaginal fornix

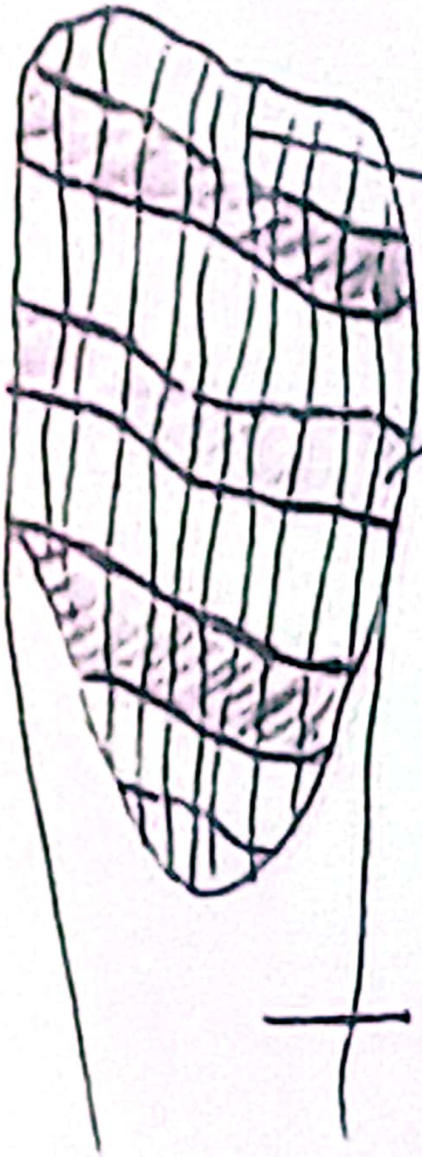
Cervix of mare



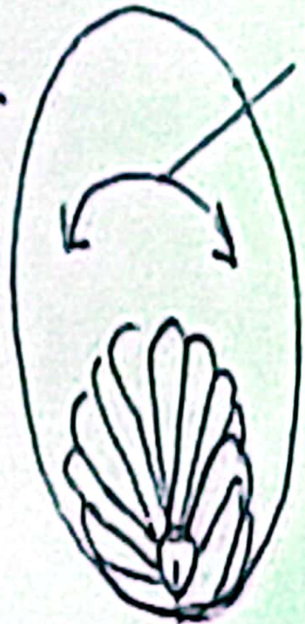
[illegible]
[illegible]

[illegible] and continuous
[illegible] . [illegible]
[illegible] vaginal speculum.

Covered in [illegible],



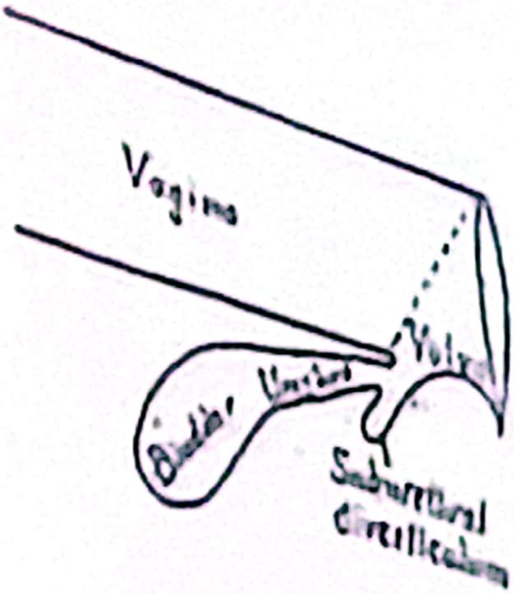
opened cervix,



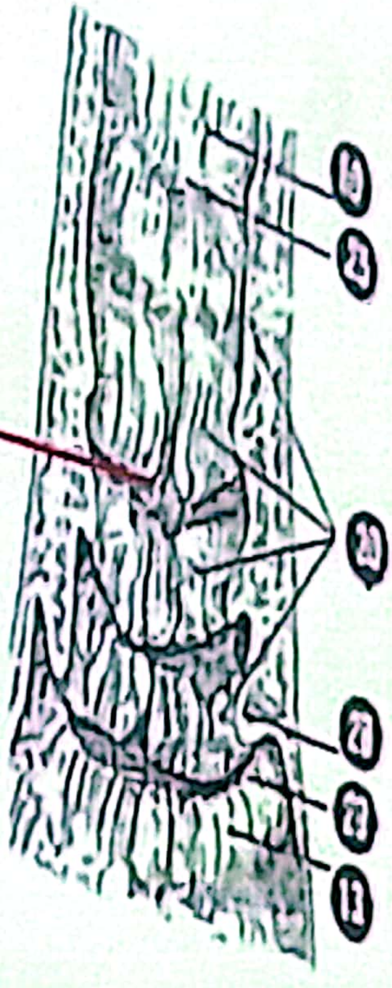
Cervix of Cow

- Fornix is dorsally only
- portio vaginalis uteri is rosette shaped

" cervix as seen through a vaginal speculum "

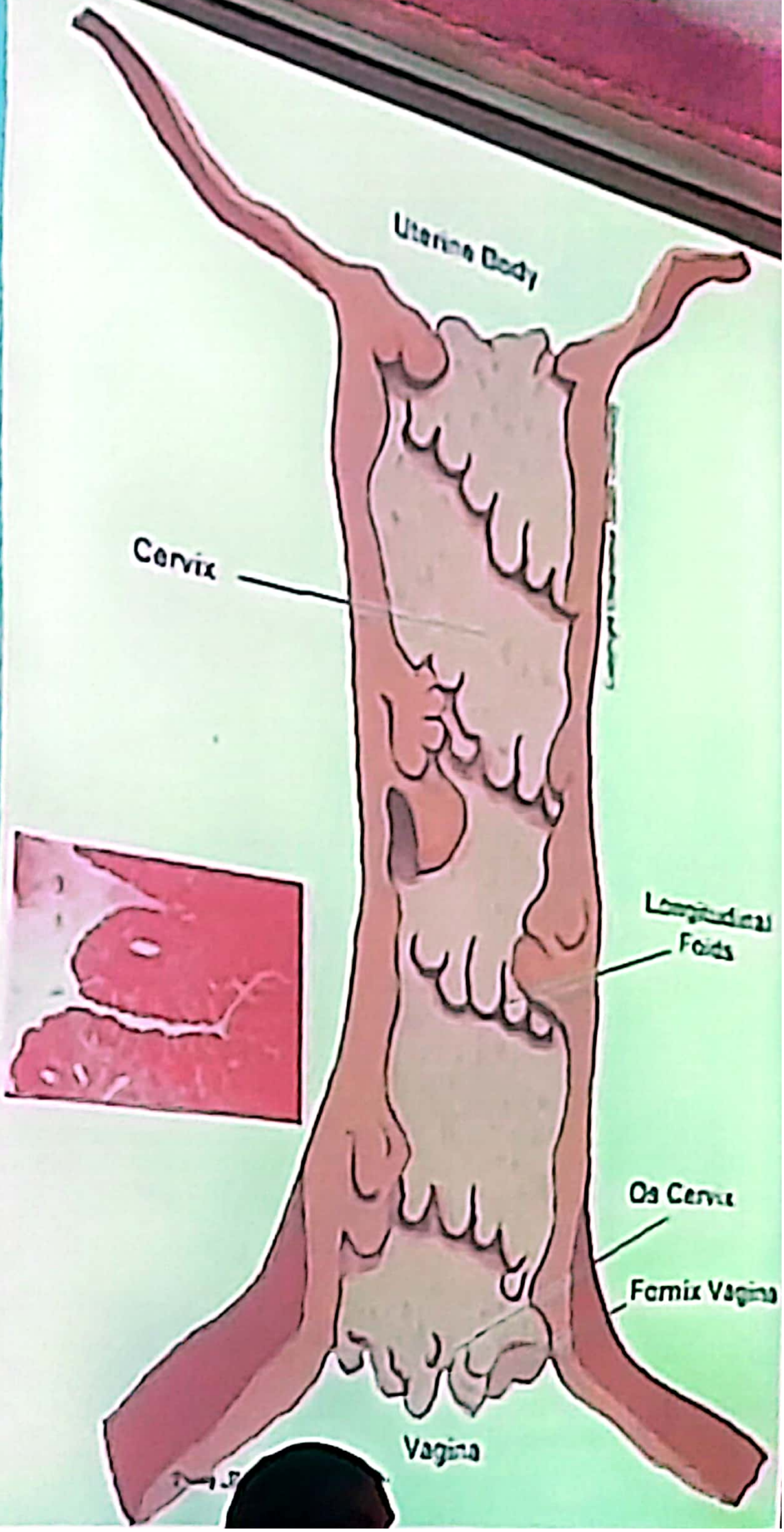


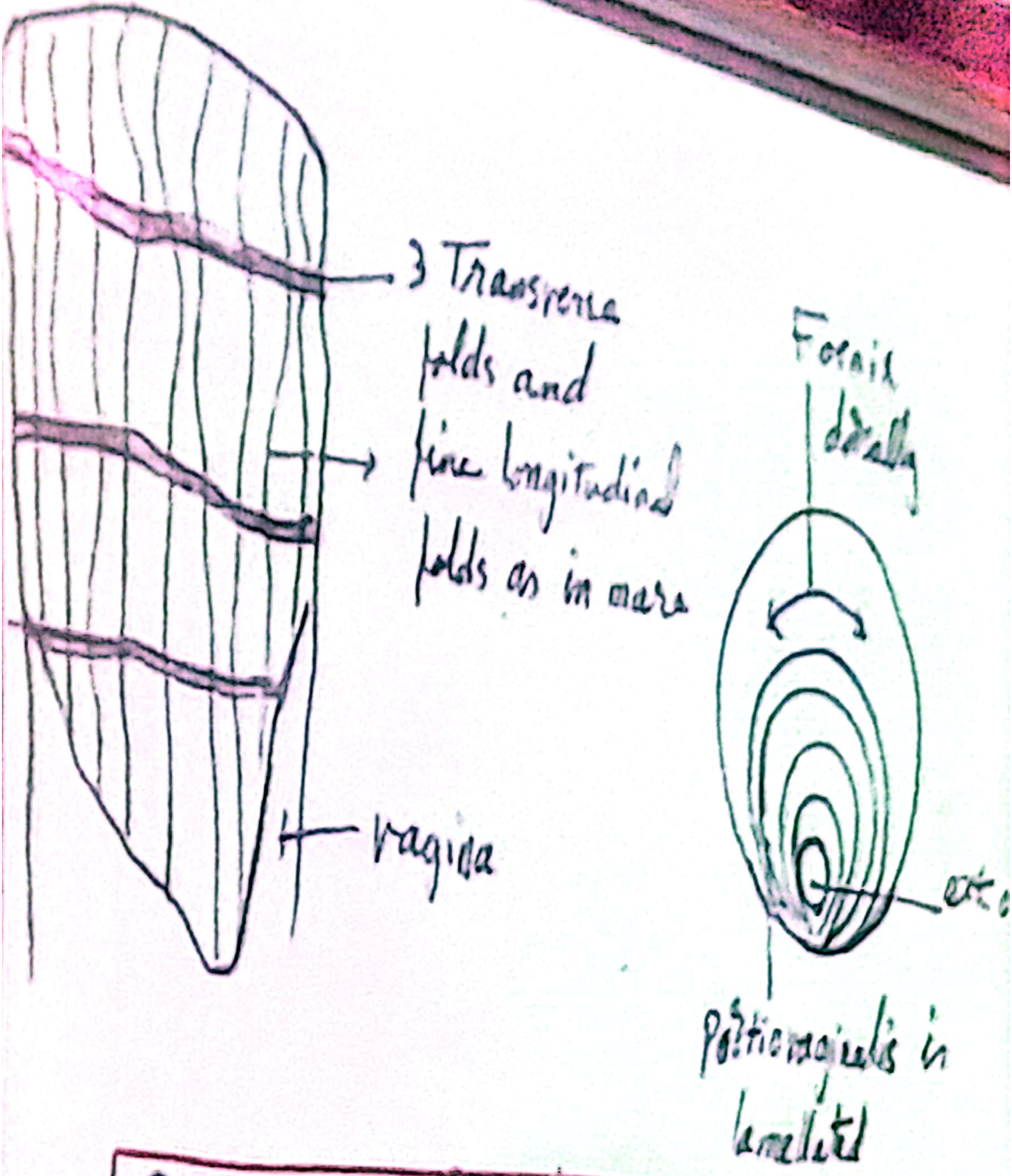
Strong Transverse and longitudinal folds



Cervix of cow

Cervix of cow





Cervix of She-Camel

vagina=female copulatory organ

*It is a thin wall tube present in the pelvic cavity, dorsally related to the rectum and ventrally to the urinary bladder.

Most o f the vagina lies retroperitoneally (caudally), and only short part is peritoneally (cranially)

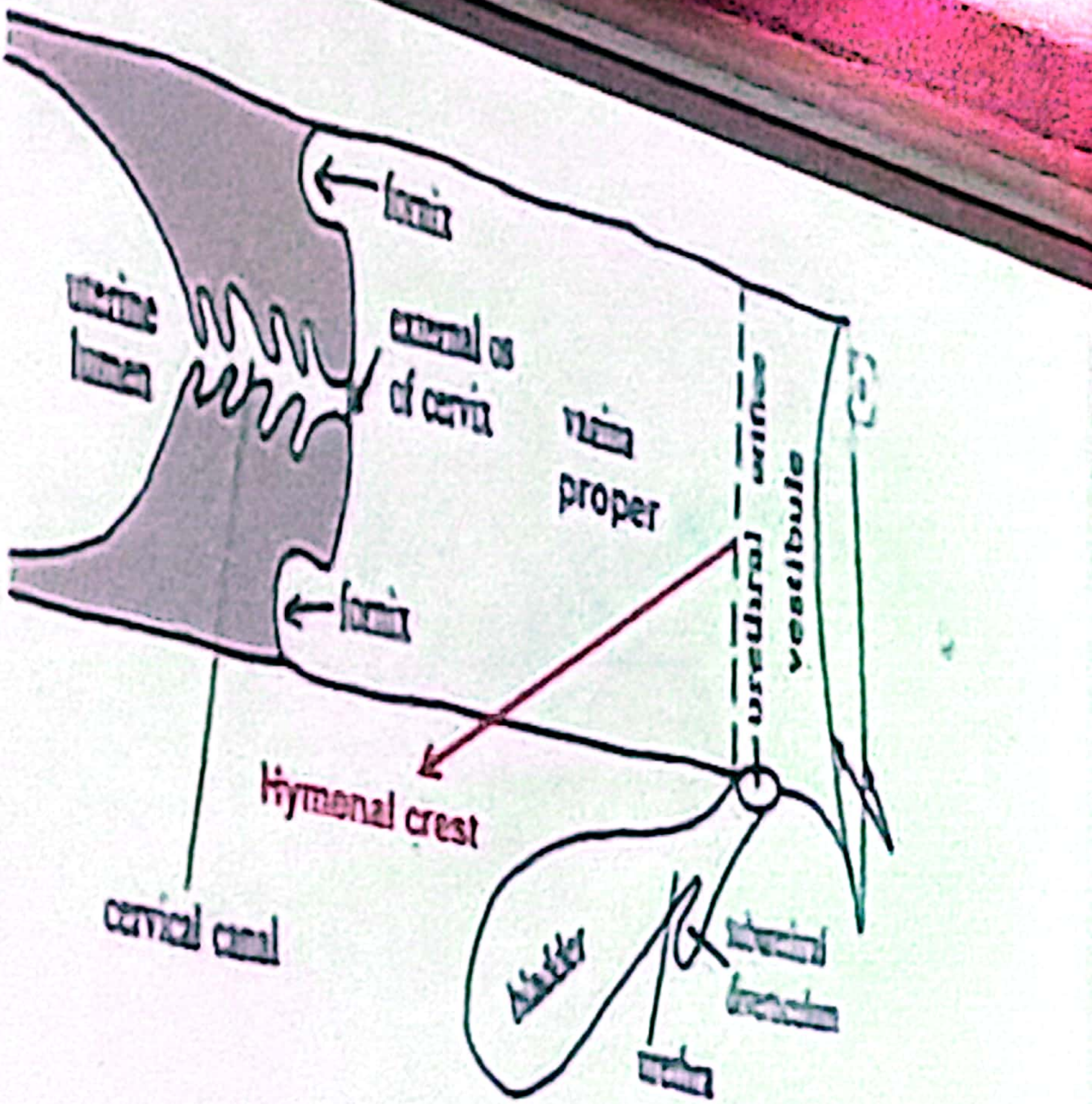
*It consists of two parts, vaginal proper extend from portiovaginalis till the external urethral orifice or hymenal crest which well developed in mare

*the vaginal vestibule, extends from external urethral orifice till the vulval lips, its wall contain vestibular glands

*in case of COW under the external urethral orifice there is suburethral diverticulum=short blind pouch.

Vestibular bulb

The wall of vaginal vestibule contain a well developed rounded venous plexus which is termed vestibular bulb ,It is well developed in mare and bitch and forms a complex erectile body .



COW Schematic illustration of caudal portion of a bovine female genital tract to show the location of the suburethral diverticulum. Arrows on the right = lips of vulva. Labeled structures: uterine lumen, cervical canal, fornix, external os of cervix, vagina, bladder, urethra, suburethral diverticulum, urethral orifice, vestibule.

The vaginal vestibule =urogenital sinus

s lined by stratified squamous epithelium and contains mucous-producing glands; the minor vestibular glands presents as solitary glands in the lateral and ventral wall of the vestibule by small orifices in linear series in case of bitch, sheep and mare , and the major vestibular glands are absent in bitch, sheep and goat ,while in cow it is located on the lateral wall of the vestibule.

In cow ,buffalo and she-camel the vestibular vagina is short, and the external urethral orifice can be often seen when the labii opened. The suburethral diverticulum is below the external urethral orifice must be avoided when catheterizing these animals.

*Gartner's ducts present on each side of the external urethral orifice

The vaginal proper

It is Musculo membranous highly dilatable canal extending from the cervix to vaginal vestibule.

The cervix projects into the proper vagina forming the portiovaginalis surrounded by the fornix vaginae.

NB the caudal part of the vagina is retroperitoneal surrounded by loose fibroelastic tissue , venous ple[illegible]

Vulva and clitoris

The vulva is consisting of two vulval lips ,right and left labia vulvas which meet at dorsal and ventral commissures separated in between by vulval cleft (vertically).

*the dorsal commissure is rounded while the ventral commissure of the vulva is pointed in all animals except mare , buffalo.

*in mare and buffalo the ventral commissures is rounded

*in she camel the 2 commissures are acute.

The ventral commissure is containing the clitoris = female penis without urethra which is well developed in case of equine, it is found in fossa clitoridis.

Parts of clitoris:-

a- crura clitoridis = two crura attached to ischial arch

b- body of the clitoris = erectile tissue called corpus cavernosum clitoris

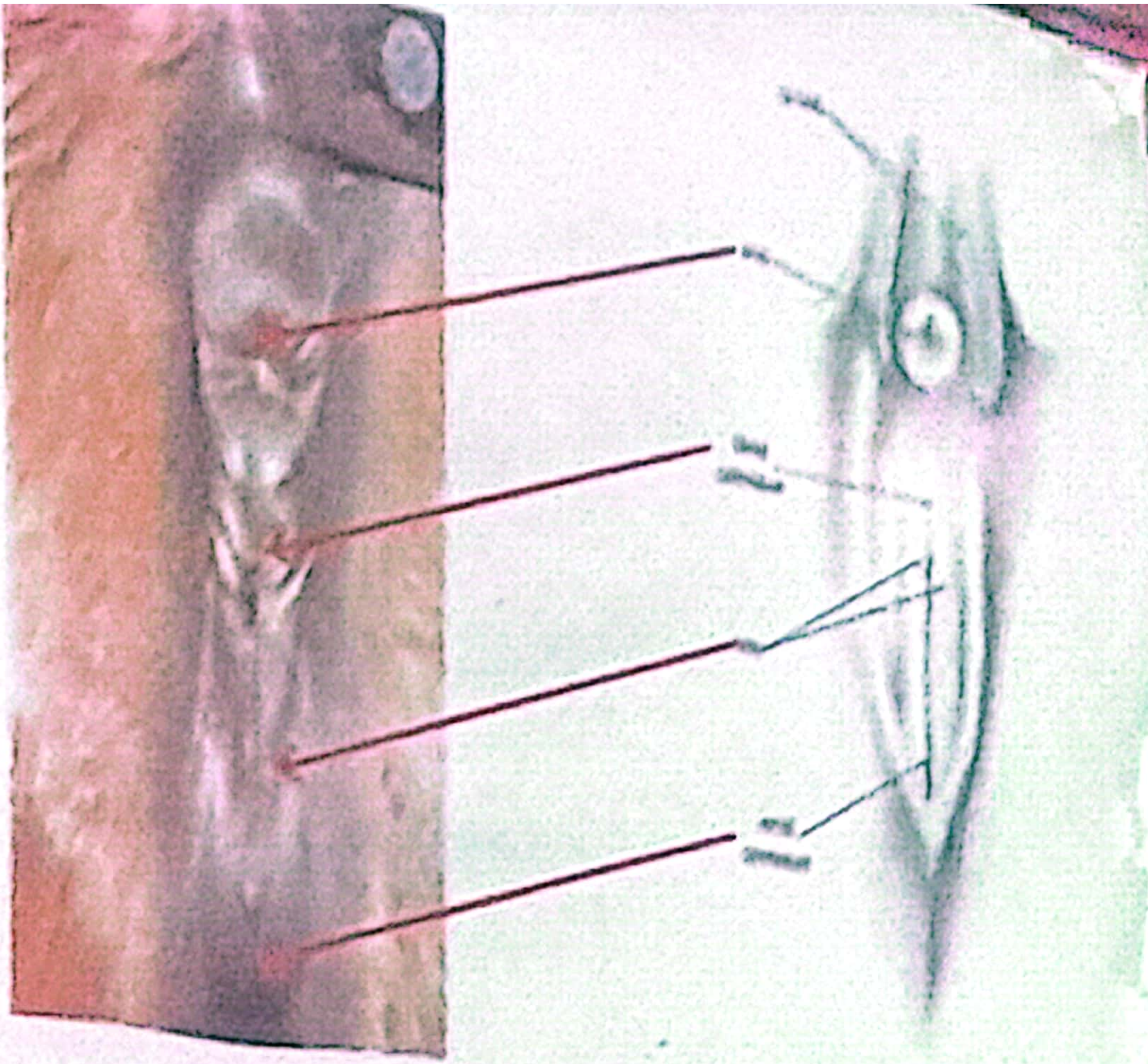
c- Glans clitoris = free end of clitoris surrounded by prepuce=preputial clitoridis.

Muscles of female copulatory organ= M. erector clitoridis

1- constrictor vestibule

2-constrictor vulvae

3-ischiourethralis

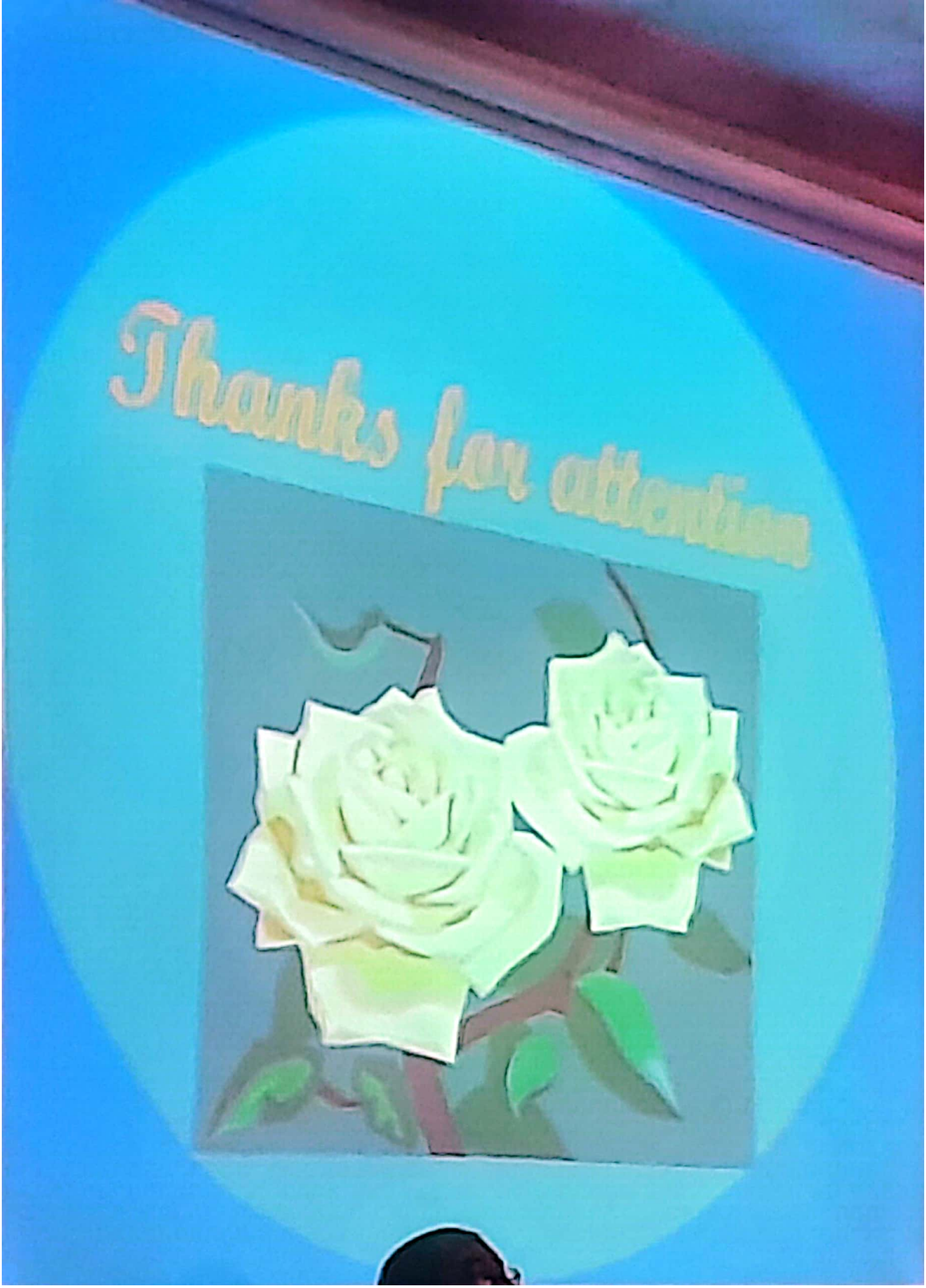


Vulva of mare

Vulva of cow

clitoris of mare

An enlargement of the (equine) vulva, showing the glans of the clitoris within the ventral commissu[illegible] ht labium; 12, glans of clitoris. TVA


Thanks for attention